NORTHEASTERN UNIVERSITY
GRADUATE SCHOOL OF COMPUTER SCIENCE
M.S. THESIS APPROVAL FORM

THESIS TITLE: Spaceship Launch: Designing a Collaborative Exergame for Familie

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M.S. Thesis Approved as an Elective towards the Master of Science Degree in Computer Science.

Thesis Advisor	4-10-14 *Date*
Thesis Reader	4/10/2014 *Date*
Thesis Reader	*Date*
Thesis Reader	*Date*

GRADUATE SCHOOL APPROVAL:

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Spaceship Launch: Designing a Collaborative Exercise Game for Families

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This is for
Noor Rahmani,
Dian Paramita, &
Siti Habibah;
for providing
a loving home
and a courageous spirit.

Acknowledgements

This thesis encompass the field of human-computer interaction, game design, health sciences, and educational technology. It would have not been possible to complete this work without the inspiration and support from my family, friends, and mentors.

I would like to express my gratitude to my advisor Dr. Andrea Parker whose compassion to people and commitment to empower underserved populations have guided my research work. Your scientific rigor and preciseness refined my contribution to science and transformed my thinking in evaluating human and social phenomena. To Dr. Magy Seif El-Nasr, I would like to thank you for your insights in game design and your feedback on research methods that helped me solidify this thesis. I would also like to thank Dr. Carmen Sceppa, Dr. Jessica Hoffman, and Cathy Wirth for guiding my human-computer interaction research endeavor within the area of health science.

I would like to thank my research partners, Ashwini and Geeta for the spot-on critiques and encouragement. To Chloe and Johnothan, thank you for supporting Spaceship Launch deployment and providing implementation ideas. To my lab partner Farnaz, thank you for your valuable feedback. Last but not the least, I would like to thank the staff of College Computer and Information Science at Northeastern University for assisting this research project.

I also would like to thank you to the wonderful Family Gym community members and staff. It was a pleasure to work with these caring families and supportive staff. In particular, I would like to thank Sarah and Raymond for the tremendous support.

To my friends Nogi, Dina, Wulan, Grace, Shinta, Bryan and Mahesh; you made the nine-months thesis work so much fun. Special thanks to Arkha, for inspiring me with learning theories.

Finally, I would like to thank my mother Noor Rahmani for the beautiful childhood that I can always look back to when things do not work out as I expected. Thank you for making me believe in what I can do. To my father, Agus Balela, thank you being a very supportive. And to my sister Dian Paramita, thank you for your fearlessness that I always model upon.

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Chapter 1
Introduction

The prevalence of childhood obesity in United States [58,75] is posing physiological and psychological health risks for future generations. It has been linked to cardiovascular diseases [32], diabetes [43], as well as psychological health problem such as lowered self-esteem that may cause loneliness and sadness [83]. Its widespread presence is burdening children of low-education, low-income, and higher-unemployment families [78]. The report released by the Center of Disease Control and Prevention in early 2014 [57] shows that obesity remains high in the U.S., and much more common among ethnic minorities. However, there is a significant decrease of obesity among 2 to 5 years old children. This encouraging progress suggests that work on combating obesity needs to be carried on with focused efforts towards lower-income families in which obesity is widespread [78] and resources are scarce [20].

While maintaining a healthy weight could be accomplished by using simple formula of consuming healthy food and spending more time in physical activity (PA), inculcating such behavior is challenging because it is impacted by numerous interconnected factors [28,42], including influence from family and home [24,52,63].

In a study on 1,454 American families, Christakis reported that the combined video games and computer games usage among children exceeded television usage [14]. The sedentariness of video games have raised concerns about childhood obesity [93]. However, given the appeal of video games to children [54,55], the research community have found its' use for health education [59] as well as an exercise alternative in the form of exergame (exercise game) [77,86,92].

The human-computer interaction (HCI) research community have studied how technological interventions are capable to encourage physical activity (PA) by using novel exercise instruments [9,27,47,53,61,80,81] or novel visualization [5,18,45]. Some work have leveraged social influences by introducing cooperation or competition between players [90,96]. However, very few have focused on family-centered intervention for promoting PA even though research has shown that family support is the key factor that influenced physical activity [24,28,30,36,42]. More work needs to be done to understand how family-centered technology-based intervention could help families model and mutually-influence healthy behavior

1.1 Research Questions

The overarching question that I intend to answer from this research is: how can parent-child interaction in a technology-based intervention facilitate health promotion and what are the design guidelines for creating such interventions? The

following are the three research questions that I attempted to answer through this thesis work:

RQ 1: What are design opportunities for family-based technology interventions that promote physical activity? Understanding how families value and assess physical activity and their barriers to achieve healthy living helps me to formulate design guidelines for family-targeted technology intervention and answer this research question.

RQ 2: How can a family-based exergame encourage parental intention to be physically active? How can parental modelling of physical activity be supported and impacted by this exergame? By answering RQ 1, I concluded that most parents who valued the health benefits of physical activity are likely to provide support for their children to be physically active. However, the interview, survey, internal observation, and BMI data of our participants suggest that the parents did not provide PA modelling even though they are actively providing support. I concluded that family-centered technology-based intervention should encourage both the parent and the kid to be physically active. Answering this question is important because research has shown that parental modelling is a strong determinant of child's physical activity level [24,30,52,63,100].

RQ 3: What types of interactions emerge in a collaborative exergame designed for intrafamily use? What factors inhibit and encourage game engagement? Answering this question is important because assessing interactions between parent and child when using this exergame prototype will provide valuable insights to refine the prototype and inform future tech-based interventions of similar nature.

1.2 Overview of Thesis

The remainder of this thesis is organized as follow. In Chapter 2, I described related work on family centered physical activity intervention as well as technology-based PA intervention including health-based consumer products available commercially.

In Chapter 3, I detail the preliminary qualitative fieldwork, which was focused on lower-income families. This demographic was selected because childhood obesity is prevalent among lower income families [78] and their resources are limited [20]. With my research partners, I assisted focus group discussions and co-conducted qualitative analysis on the interview data. I acquired data on parental attitude and support towards physical activity, as well as emotional aspect of physical activity, and how technologies—such as smartphone and video games—are regulated and integrated within the families.

Based on the finding on this formative preliminary study, I designed Spaceship Launch, an exergame deployed in family and community setting. In Chapter 4, I describe the design rationale and technical aspect of this game and the results of the field study I conducted to evaluate the system. I detail the various

methods (interviews, participatory design workshop, surveys, and log analysis) that I used to evaluate the Spaceship Launch. Then I conclude with the result of the evaluation.

In chapter 5, I reflect back to previous work and the result of the Spaceship Launch evaluation. Then I synthesize the findings into a framework for designing family-centered technology-based interventions in the form of exergames or gamified systems. This informs the direction of future research on the value of family in fulfilling the psychological needs that make exergames enjoyable and motivating. The findings also highlights that technology-based interventions should be designed as a learning experience with specific learning outcomes.

Finally in chapter 6, I summarize the themes that emerged on my study and how my findings answer my research questions. Then I conclude with future research directions that could be guided by this work.

1.3 Contributions

My work is a contribution to HCI in four ways. First, through the design and evaluation of Spaceship Launch, I provided a case study on the use of exergames for promoting physical activity among lower-income families in United States. Second, through this case study, I highlighted how family could impact the enjoyment and motivation of exergame. I also highlighted how exergame experience could also be designed as a learning experience with a defined set of outcomes. Third, from the result of the case study, I provided a preliminary design framework to inform future work on technology-based PA intervention. Finally, based upon my research, I present a set of recommendations for future research on HCI and health promotion.

Chapter 2
Related Work

This research is primarily guided by previous works on health technologies in the field of HCI, Computer-Supported Cooperative Work (CSCW), and Ubiquitous Computing (Ubicomp). I am also motivated by public health research in addressing childhood obesity using family and community centered approach. This chapter will start with an overview on family centered approach. Then I will explore works from HCI research community on the use of health technologies for physical activity promotion as well as consumer products with similar purpose.

2.1 Family–Centered Physical Activity Promotion

Childhood obesity has been linked to numerous adverse health risks, such as cardiovascular diseases [32] and diabetes [43]. Study by Strauss suggests lowered self-esteem among obese children which may cause increased level of loneliness, sadness, and nervousness [83]. Children with lowered self-esteem are more likely to smoke and drink alcohol [83]. Obesity is also much more prevalent among children of low-income families [78]. In early 2014, the Center of Disease Control and Prevention reported that obesity is still alarmingly high in United States, affecting 34.9% of the population. While the rate of childhood of obesity has plateaued since 2003-2004, its prevalence remains high with 16.9% of children aged 2-19 reported to be obese. One encouraging news is the number of obese children aged 2-5 has decreased substantially from 13.9% in 2003-2004 to 8.4% in 2011-2012. This suggests that work on combating obesity has shown a promising result and needs to be carried on.

Health scientists have indicated that obesity problem in United States are beyond personal health issue and have looked into changes at the environmental level to reduce obesity [36]. Ebbeling classified home and family environment as one of the area that needs to be targeted to cure obesity epidemic, along with school environment, physical environment, and public policies [28]. Similarly, Kumanyika highlighted home and family setting as one of the five factors that induces childhood obesity [42]. Other factors includes: media and marketing, food access and availability, built environment, and school setting [42].

Indeed, the role of family has been described to be "a powerful and lasting determinant of preventive health behavior" [72] as it is a source of modelling, influence, and support [88]. A ten-year research outcome provides support for family-based treatment for combating childhood obesity: a higher rate of success present when parents are involved in the treatment [30]. Family support also has influential effect on adults. In a study by Wankel, while most adult participants are motivated to join exercise program for the health benefits, it was the family support that differentiate continuing participants with the dropouts [94]. Grimes ET. al. [35]

also highlighted how collaborative health reflections allows parents to model positive behaviors to their children, and at the same time allows parents to receive motivating feedback from their children.

The role of parental modelling has been cited as one of the predictor of later exercise participation [24] and this claim has been supported by a number of research [37,100]. Moore reported that when both parents are active, 68% of the children in the study were more likely to be active; and this likelihood is lowered to 27% when both parents are not active [52]. The amount of time parents spent in physical activity is also significantly correlated to children physical activity pattern [63].

The impact of parental modelling may be diminished if the child does not directly see their parent's activeness (i.e. the child and the parent did not exercise together). This is suggested by several studies that found child physical activity did not increase when the child did not see their parents being physically active [23,91]. In the study by Trost et. al. in parental influence on youth physical activity, they found that parental modelling would not boost child's self efficacy in PA [91]. Trost argued that this is attributed to the limitation of his study where parental modelling was measured as child's perception of parental PA level, as opposed to actual parental PA measurement. This suggests the value of having children to realize how active their parents are. Thompson further elaborated the challenge of providing parental modelling by providing accounts how families are facing a number barriers to exercise together [89]. These barriers are described to be time barrier, financial barrier, bad weather, lack of access to facilities, and varying interest among family members. Thompson concluded by stating "family-based interventions might be more effective if they are designed to accommodate the complex demands and needs of two-parent and single-parent families" [89].

2.2 Social Cognitive Theory

In the previous section I described that family factor has been shown to be a dominant factor in health behavior change, as it is a source of modelling, influence, and support. Peer and environmental role in behavioral change has formed a set of concepts that build the Social Cognitive Theory (SCT). SCT works on the concept that individuals adopt new behavior in part through self-regulation and by observing others.

Observational learning, the capacity to learn form others, is a central tenet of SCT. Access to family and peer determines what behaviors one can observe, which reinforces the value of parental modeling in inducing PA behavior. Family and friend modelling is found to be a strong predictor of PA [23] and this is confirmed by a literature review by Dishman et. al. [24]. Sallis speculated that ethnic minority are more likely to turn to families as a source of modelling [74], reiterating the value of using SCT as a framework when targeting intervention to this group.

Behavior change in SCT is predicted by *outcome expectation* and *self-efficacy*. *Outcome expectation* is described as individual's belief whether certain outcome could

be attained if the individual chooses to perform the perceived necessary actions. This helps to explain why some individuals choose actions that may not have immediate rewards in the interest of longer-term benefits. *Self-efficacy* is described as a person's judgement of his/her capacity to complete tasks and accomplish goals, and is known to be a predictor of PA participation [26].

Outcome expectation is mediated by *self-regulation* [39] which puts emphasis on concrete skills for managing oneself [33]. Glanz [33] laid out six ways that self-regulation can be achieved (summarizing principles originally proposed by Bandura [8]): self-monitoring, goal-setting, feedback, self-reward, self-instruction, and enlistment of social support.

2.3 Self-Determination Theory

The Social-Cognitive Theory (SCT) provided a framework that explains how families could model and influence physical activity behavior. However, SCT do not directly explain individuals' motivations to be physically active. Deci and Ryan [71] formulated the Self-Determination Theory (SDT) which explains that individuals' motivation are diverse and fostered by the satisfaction of three psychological needs: *autonomy*, *competence*, and *relatedness*. The notion of self-determination does not disregard the value of peers, as relatedness is associated with motivational environment from parents, teacher, or peers.

Deci and Ryan's description of motivation continuum and how motivation is regulated is shown in Figure 1 [71]. *Amotivation* is the state of no intention to act. At this state, individuals either do not act or act without any intention. To the right is *External Motivation* in which individual's actions are driven by external regulations. The least self determined is *external regulation*, in which individual's actions is determined by external causes or rewards. The next regulation is *introjected regulation*, in which actions are motivated to avoid the feeling of guilt. The following is *identified regulation*, in which individuals are motivated by the benefits resulting from the actions. The most determined type of regulation is *integrated regulation*, in which the introjected and identified benefits have been assimilated too individual's self-regulation. *Integrated regulation* is still considered as externally motivated, because motivation is still driven by external factors. Finally, the most determined individuals are those with *Internal Motivation.* Individuals of this type are motivated for the intrinsic enjoyment generated by the activity itself.

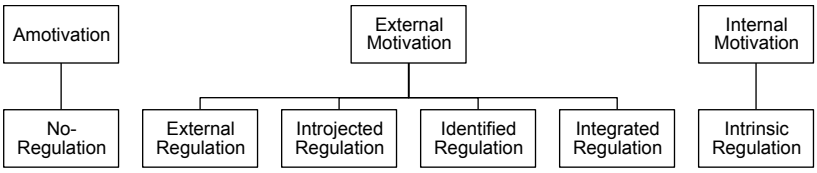

Figure 1 Deci and Ryan's motivation types with their regulatory styles

This continuum of motivations is determined by three psychological needs [71]. The need of *autonomy* is described as one's perceived control to accomplish a goal. Autonomy is not associated with individuality or independence, but rather the feeling of capability to complete a task whether it as an individual or with other people. External reward can have a negative outcome as it lowers the sense of autonomy, because the feeling of control over the outcome is affected by external reward. The need of *competence* is described as individual's perceived capacity to complete a task. The third and last factor is the need of *relatedness*, which is described as the need to feel connected with others and be concerned for them. Relatedness is also related to the need to feel accepted in the social environment. For example, team sport players build strong relationship with their teammates, which boost their effort to win [62].

Murcia ET. al. [6] studied the relationship of autonomy, competence, and relatedness with physical activity enjoyment and commitment. Specifically, Murcia looked into the two types of motivational environment: task-involved motivational climate (i.e. the focus is on task domain and personal improvement) and ego-involved motivational climate (i.e. the focus is on comparisons with others). The result shows that task-involved climate positively predicted the fulfillment three psychological needs: *autonomy*, *competence*, and *relatedness*; whereas ego-involved climate predicted none of those. Murcia reasoned that motivational climate that put emphasis on cooperation, effort, and personal improvement leads exerciser to feel they are more competent since it is "a self referenced criteria, which is far more controllable" [6].

The SDT provided me a framework to understand how task climate affects individual's motivation in PA as well as factors that foster engagement.

2.4 Serious Games and Gamification

The research community have explored the use of computer games for health education [59] and for physical activity alternative in the form of exercise game (or exergame in short) [77,86,92]. In this context, games are no longer serving its' original purpose to entertain but also to deliver a larger and a more serious purpose. This results in a new game genre called "serious game" which was defined by Ritterfeld et. al. as "any interactive computer-based game that has been developed with the intention to be more than entertainment" [66].

Serious games should not be confused with *gamification*. Deterding defined gamification as "the use of game design elements in a non gaming context" [22]. Deterding further made the distinction between the two, defining *serious games* as "full-fledged games for non-entertainment purposes". Deterding also distinguished *gameful design elements* from *playful elements*, by arguing that *play* is free-form, expressive, and improvisational. In contrast, *games* are structured by clear rules and promoted by goals or competition.

These distinctions, however, elucidated one unifying outcome of serious games and gamified systems: the non-entertainment purposes. For the lack of better term, the gameful element and playful elements could fall into the category of entertainment purpose. These two elements are designed to produce the non-entertainment outcome (Figure 2).

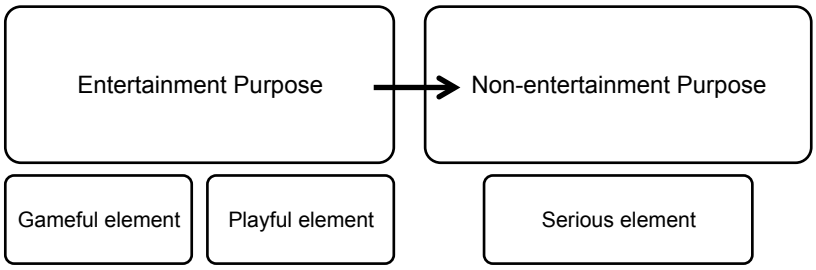

Figure 2 Relationship between gameful, playful, and seriousness

Deterding further identified what constitutes as game design elements and organized these elements into levels, ordered from the most concrete to the most abstract (Table 1).

Level	Description	Example
1. Game interface design patterns	Common, successful interaction design components and design solutions for a known problem in a context, including prototypical implementations	Badge, leaderboard, level
2. Game design patterns and mechanics	Commonly reoccurring parts of the design of a game that concern gameplay	Time constraint, limited resources, turns
3. Game design principles and heuristics	Evaluative guidelines to approach a design problem or analyze a given design solution	Enduring play, clear goals, variety of game styles
4. Game models	Conceptual models of the components of games or game experience	MDA; challenge, fantasy, curiosity; game design atoms; CEGE
5. Game design methods	Game design-specific practices and processes	Playtesting, playcentric design, value conscious game design

Table 1 Game design elements identified by Deterding

To understand the playful aspect of user experience, Arrasvuori ET. at. developed Playfulness Experience (PLEX) framework [7]. This framework consists of 22 categories as shown in Table 2.

Playful element	Description
Captivation	Forgetting one's surroundings
Challenge	Testing abilities in a demanding task
Competition	Contest with oneself or an opponent
Completion	Finishing a major task, closure
Control	Dominating, commanding, regulating
Cruelty	Causing mental or physical pain
Discovery	Finding something new or unknown
Eroticism	A sexually arousing experience
Exploration	Investigating an object or situation
Expression	Manifesting oneself creatively
Fantasy	An imagined experience
Fellowship	Friendship, communality or intimacy
Humor	Fun, joy, amusement, jokes, gags
Nurture	Taking care of oneself or others
Relaxation	Relief from bodily or mental work
Sensation	Excitement by stimulating senses
Simulation	An imitation of everyday life
Submission	Being part of a larger structure
Subversion	Breaking social rules and norms
Suffering	Experience of loss, frustration, anger
Sympathy	Sharing emotional feelings
Thrill	Excitement derived from risk, danger

Table 2 PLEX framework categories

Identifying the concepts that constitutes a serious game or a gamified system is useful to analyze the negative outcomes of these systems. Ratan and Ritterfeld argued that the term "serious game" suggests the game's outcomes should always be advantageous to the player: serious game should not have any negative result, and should always delivered a sustained learning outcome [65].

One aspect of games that has been suggested to cause negative effect is competition. Work by Adachi suggests that competition, not violence, may be the characteristic of a video game that has the greatest influence on aggressive behavior [1]. Zagal bring up other types of game interaction as the alternatives to competition: cooperation and collaboration [99]. Two similar works by Rocha and Smith fleshed out the distinction between competition, cooperation, and collaboration and its implication to user interactions [68,79]. In the next two paragraphs I will summarize the categorization from this studies.

A *competitive game* is described as a zero-sum game in which the outcome is always one player wins and the other loses. Thus the challenge is to outdo the other player. In a *cooperative game*, the game mechanics require two players to work together to achieve a *collective goal*, even if the individual goals may be different. However, in a cooperative game one player could betray the other player. A game could be cooperative at one point and competitive at some point. Two players could initially work together to complete a collective goal and become competitive later in

the game. One game mechanic that affords cooperation is complementary powers, in which the collective goal could only be accomplished if players with two different powers cooperated. The other mechanic is team problem, in which the completion of a collective goal needs more than one player.

In contrast, a *collaborative game* would have an intrinsic mechanics that guarantees players who worked together to complete a collective goal will always get an equal benefit. If one player loses, then the other player will lose.

Collective goal is a common theme across cooperative and collaborative. A study on 4 cooperative games by El-Nasr [76] found that collective goal encourages players to help each other, work-out problem together, and laugh together. This finding motivates my work to implement cooperative or collaborative game as part of my intervention design for families.

2.5 Technology-Based Physical Activity Promotion

The HCI research community has explored the use of interactive computing to promote physical activity. I categorized this technology-based intervention into three categories: (1) exercise game or exergame; (2) novel visualization; and (3) social-support technologies. After I describe the current state of research in this area, I proceed with a number of health products available in the consumer electronic market.

2.5.1 Exercise Game (Exergame)

One of the research in exergame for children is iFitQuest, targeted for adolescents aged 11-12 [46]. IFitQuest consists of two mini-games with open-ended goal: "Escape the Ghost" and "Collect the Coins". "Escape the Ghost" is sprint game in which players must sprint faster than a predetermined time to avoid being caught by the virtual ghost. Level are automatically adjusted when the player wins on loses. "Collect the Coins" stimulated short sharp bursts found in agility training exercise. The game player must sprint toward certain direction, and after the said location has been reached, the game provided another direction to sprint, and so on until all the coins have been collected. Players are free to choose which game and level to play.

At the end of the 7-day field trial, Macvean highlighted the presence of "novelty effect", where the game's effectiveness to facilitate PA was eventually wearing off [47]. While initially the participants was reported to spent 60% minutes of their time in moderate to vigorous physical activity, beginning from the 4th week the number fell to 35%. "Novelty effect" was also reported on Fish'N'Steps [45] and American Horse Power Challenge [64] which I will discuss later in this chapter.

Macvean's work is also important in identifying exergame player's motivation and the role of competition in an exergame [47]. Some players are motivated by *Big-picture Competition* to accrue most points by beating other players in the game. Other players are motivated by *Inter-competition* with selected players. *Mastery*, motivation

to beat self-set target, was also present on the field study. Other player preferred to use the game away from the peers, suggesting the negative impact competition.

Similar to MacVean's work, the American House Power Challenge (AHPC) exergame was also deployed in school setting. This large scale game was deployed in multiple schools and the game objective is to encourage students to collect as many steps as possible to beat other schools [29]. Xu et. al. identified five player types emerging from this game: Achievers, Active Buddies, Social-Experience Seekers, Team Players, and Freeloaders [96]. Some of these player types parallels to Macvean's work I discussed earlier [47]. Achiever seems to corresponds to players who are *Mastery* motivated and Active Buddies mirrors the characteristics of players who are motivated to *Inter-competition*. Team-Player shows similar motivation to players who are motivated by *Big-Picture competition*, although Team-players are motivated to beat other schools rather than other individuals. Social-Experience Seekers and Freeloader are categorized by Xu as players who did not engage to the physical activity aspect of AHPC [96]. However Social-Experience Seekers are still engaged to the social aspect of the game, whereas Freeloaders are not engaged to the game at all.

Over the 11 month deployment of AHPC, Xu also identified the presence of "novelty effect" [97]. The number of steps tapered dramatically at the end of the deployment. Reduced interest, technical glitches, schedule mismatch, and weather change was cited as the reasons of the low engagement. AHPC's reliance to administering figure (i.e. teachers) to maintain engagement to a game at this scale was identified by work of Poole [64].

PiNiZoRo (stands for Pirate Ninja Zombie Robot) is a GPS exergame targeted for family setting [81]. Similar to conventional treasure map game, parents set up "battle locations" using GPS points on their neighborhood. Using a smart-phone, children navigate the digital treasure map to find the "battle locations". Finding a location will activate a mini game, which the child can play. Participants on PiNiZoRo field study appreciate the game as a motivational tool to exercise more.

Some exergame implemented novel extortion device. Using a ball with electronic sensors, UbiBall combines obstacle course game with screen-based computer game [27]. The objective of this game is to accomplish as many game points as possible in the real world to gain power-ups in the virtual game. While the children playing this game unanimously thought the game was exciting and fun, it was unknown if they understood the relationship between the real-world game and the virtual game. Similar work by Adillon used conventional air sliders with projected game interface [80]. The projected game interface allows extortion level to be controlled. Other research in exergame includes Move2Play [10], Wi-Fi Treasure Hunt [15], and ball-throwing game for early age children [61].

Several games I discussed in this section falls into the category of casual games, which is characterized as a game with simple controls and low learning barrier

[69]. In the next section I will discuss two exergames with a more sophisticated control and requiring more skill to master.

Unlike exergames in conventional term, in which exercise is an intrinsic feature of the game, "Play, Mate!" integrates non-coercively to an existing game (Neverball) by encouraging player to be physically active to get extra game time [9]. The game was modified to reduce the default time to accomplish the game task and player was given the opportunity to get extra time by performing jumps. Jumps were tracked using an external activity monitor worn on the waist. Players using the modified game reported similar game enjoyment level to players using the unmodified game, even though they were eight times more engaged in physical activity than players on the unmodified game.

SpaPlay is a full-fledged game in which PA and healthy eating is an central part of the game [25]. The objective of the game is to maintain a virtual spa resort. In order to maintain the rating of the spa, game player must perform physical activities and make healthy eating choices. Game player can either do sparks (single short task) or quests (group of task that takes few days to a week to complete). Completing sparks and quest will unlock new contents that improve the aesthethics and rating of the resort; whilst infrequent engagement will cause the resort's aesthetic to be poorly affected and rating lowered. In a 45 days field study of 18 people, it was found that the participants eventually integrates the SpaPlay sparks and quests to fit into their existing routine [25], suggesting that the game eventually become a source of reward of the participants' existing routine.

Exergames either could fall into serious games or gamified systems depending on the completeness of the game design elements. However, it is more important to identify the gameful, playful, and serious elements of an exergame. This is to see the relationship between the three and how they are affecting one another. The presence of novelty effect in iFitQuest could be described as the diminishing value of *gameful* and *playful* design elements that are no longer able to sustain the *serious* purpose of exergame. The emerging competition types in exergames could be described as the use of *gameful* and *playful* design elements that produces user interactions that promotes and inhibits the desired *serious* outcome. These concern guided my intervention design that will be discussed in the concluding sections.

2.5.2 Novel visualization

At least six HCI studies have been conducted to see how personal PA visualization affected people [4,5,17,18,44,45]. Fish'N'Steps links users' steps count to the growth and happiness of virtual fishes in a virtual aquarium [45]. Steps count is captured by taking the picture of the pedometers screen. In the field study, Fish'N'Steps participants were asked to wear the pedometers for 4 weeks, then participate in the game for 6 weeks, and wear the pedometers for an additional 4 weeks for post-intervention measures. Using Trans-theoretical Model (TTM) as a framework, participants' stage of change was measured during pre and post

treatment. Greater leap of stage of change was evident among participants in pre-contemplative stage. The visualization had little effect among participants in the preparation, action, and maintenance stage, but the game nevertheless provides additional motivation to be physically active.

The virtual fish aquarium was reported to provide positive as well as negative incentives [45]. Some participants are happy to see their fish grow, whereas other avoided seeing the sad fish when the participant was not active enough. Engagement with the virtual fishes seems to be poor for lowest and highest stage of change: participants with the lowest levels were discouraged by the overly negative state of their fish, whereas participants with the highest stage of change is not affected by the visualization. Lin also found the competitive aspect of Fish'N'Steps has discouraging effect on some people and reported the presence of "novelty effect" where participants perceive the game as increasingly repetitive over time [45].

A similar visualization is UbiFit Garden, where flower and plants in virtual garden was used to visualize physical activity level. However, unlike Fish'N'Steps where steps count was entered manually, UbiFit Garden used an external activity tracker that automatically feeds the data to a glanceable visualization on a smartphone [18]. On a 3-week field trial, participants reported that the visualization is motivating and helped them to focus on planning physical activity. However, the device's inaccuracy to infer the correct physical activity pattern was reported to cause disappointment to some user.

Fish'N'Steps and UbiFit exemplified the gamification of physical activity data. In these systems, PA data was gamified using *gameful* reward system and *playful* designs (in the form of fishes and flowers) to serve its *serious* purpose (raising PA awareness).

Findings from this work detailed how *gameful* and *playful* design elements may affect the desired outcome. In UbiFit Garden, the perceived innacurate sensor illustrated how the value of gameful reward system may diminish if the underlying system is not robust. Fish'N'Steps implemented a leaderboard game design element that lead to competition and in effect demotivated some users. Fish'N'Steps also implemented the *cruelty* PLEX playful category that "punishes" users who are not active by making their fish to look sad. This is shown to have negative outcome on some people. Novelty effect is also present in Fish'N'Steps. This is another instance when the appeal of *gameful* and *playful* design elements wears off and limits the system's ability to serve its intended serious purpose. The relationships between *gameful, playful,* and *seriousness* further guided my intervention design decisions.

PA visualization on it's own have been studied to understand how providing contextual information with quantitative PA measures (*i.e.* further information about the physical activity, such as where it occurred) can help people to find physical activity opportunities and reasons why one activity is more suitable than another [44]. Using three studies, Li ET. al. explored the role of contextual information in physical activity awareness system. In the initial study, participants were asked to log

their activity manually on paper journal. The participants were supplemented with physical activity data from the wearable tracker. In the second study, participants logged their physical activity data and contextual information (activity done, its location, and people participating in the activity) on a computer system and were given a visualization of their physical activity level. Participants reported increased PA awareness after the second study, and contextual information helped them to identify PA opportunities that they are not aware of. This finding suggests that associating physical activity with contextual information has the potential benefit of increasing the awareness of PA opportunities.

In the third and final study, Li's participants' physical activity and contextual information (GPS location) logging was fully automated using a smartphone and external GPS device. In contrast to the second study, participants was reported to be less aware of PA opportunities after using the fully automated logging. Li proposed a compelling argument to explain this: "taking away too much of the responsibility of monitoring may have a detrimental effect on immediate [PA] awareness", and proposed to balance the reduced engagement with "more engagement during the reflection on the data" [44].

In the past couple of years, health monitoring devices have been available for consumer market presenting a feature to monitor PA level similar to the work of Lin, Consolvo, and Li. Fitbit offers wearable activity trackers in various form factors and provides a dashboard to helps users assess their physical activity level. Fitbit's dashboard offers time-series graphs in which user could see their steps and calorie expenditure in 15-minutes granularity. The system allows user to set their steps, calorie, distance, or minutes goal. On Fitbit Fit model, the wristband will provide unobtrusive notification by vibrating when the user reached the desired PA target. Based on my observation, the Fitbit dashboard is not suitable for people with low health literacy, because there is no prominent information on how these physical activities affect health. Fitbit users needs to familiar with the concept of calories, recommended weekly time in PA, as well as the complex relationship between food intake, energy expenditure in physical activity, and how it affected a healthy weight.

Similar wearable tracking device for consumer market is Nike Fuelband and Jawbone Up. While both offer physical activity graphs similar to Fitbit, Nike's Fuelband is unique as it introduces the concept of "Fuel" which acts as a proxy unit to help user understand physical activity level.

2.5.3 Social-support technologies

Technology has been used to provide social support by enabling new forms of interactions that was previously unattainable. "Jogging the Distance" explores how technology could be used to accommodate the challenge to find exercise partners to gain social support [56]. Thompson highlighted that similar challenge also apparent in family-centered PA interventions [89]. In "Jogging the Distance", jogging partners who could not be at the same geographical place communicated over the hands free

mobile phone. While jogging in conventional sense requires partners to have matching pace, in this form interaction the pace of each jogger become less relevant. Participants reported that being able to communicate was more important and only 16% reported a strong desire to compare the pace of their partner. Having an exercise partner was described as "far better than running alone".

Mueller extended this concept in "Breakout for Two", an interactive large display system where two people who are miles apart can play moderate/vigorous physical activity game [53]. It was reported that virtual physical game over the distance allows them to bond with new players, compared to controlling the game using computer keyboard. However, the experience with the game was reported to be much better if played with familiar people than with strangers.

In Chick Clique, participants formed a group of four in which they collaboratively collect steps [90] and engage in a friendly competition among participants to raise physical activity level and health awareness. Participants reported that having group performance rated by the peer affected their motivation to be physically active. However, a few raised concerns that competition may lead to excessive exercise.

The impact of competition in family-centered technology-based intervention also studied in "Family Healthy Living Portal" study [16]. In this study, family members were asked to share healthy living ideas in an online we portals. Participants was divided into three groups: (1) no goal, no feedback, (2) with goal, but no feedback, and (3) with goal and comparative feedback. Families that were presented with a goal shared more healthy ideas than families who did not receive any goal. However, families with goal and received feedback (in the form of other families' achievement) shared the most ideas. The mothers seems to be the driving factor in interfamily competition, as the study found that mothers who received comparative feedback shared the most physical activity ideas. Colineau suggests that mothers are more perceptive to the competitive environment in others-referenced competition [16].

2.6 Discussion

Social Cognitive Theory (SCT), Self-Determination Theory (SDT), and the research in HCI, game design, and public health provided me a wealth of resources to theoretically-ground my family centered technology-based intervention. Research suggests that intervention at home and family setting offers a shared space where family members could provide modelling, influence, and support [88]. Additionally, physical activity intervention produced higher rate of success when deployed in a family setting. From the perspective of SCT, this emphasized the value of observational learning. However, as highlighted by Thompson [89] and raised by Pate [60], providing parental modelling is challenging because various kind of barriers prevented parents and children to be physically active together.

A potential intervention is to leverage the connective capability of computing technology in to allow parents and child to influence each other. Research in HCI have provided models on this form of interaction. In "Jogging the Distance", O'Brien provided example how technology helped two people—who are not in the same geographical place and have different jogging pace—to provide support to each other [56]. In "Chick Clique", Toscos [90] provided evidence how adolescent could support each other and be motivated even if they do not share the same physical space. These findings opens new opportunities for technology where parent and child's physical activity could potentially be modeled and influenced even if these activities do not occur at the same time or place. It is evident that almost all of the work in HCI barely touched interventions in parent and child context, thus more work needs to be done in this field.

2.6.1 Competition vs. Collaboration

Much work has cautioned the potential perils of competition. While competition has positive motivational effect on certain individuals [16], it has been reported to have demotivation effect on another [34,45,47]. Some iFitQuest players were said to avoid playing the exergame with his/her peers, which suggest negative impact of competition [47]. Fish'N'Steps participants were discouraged with the game's competition because they felt they already had too much competition in the real life [45]. Work by Grimes et. al. on health technologies to support reflections also raised parental concerns of the negative effect of comparison and competition to their children [35].

This solidifies Ratan's concern that the use of gameful element may not always produce advantageous effect [65]. There are two recommendations to balance the undesired effect of competition. Grimes recommended collaboration between family member instead of competitions [35], for example by emphasizing "how healthfully the family is living as a unit, rather than how each person is doing compared to other". Tassi proposed to shift the focus from *other-referenced competition* to *task-centered competition* [87]. Therefore, rather than competing to beat other players, an exergame should be framed to encourage families to achieve recommended PA level. Whitehead also proposed to emphasize PA on individual's mastery rather than peer's comparison. [95]. This resonates to Murcia's work on SDT, that recommends cooperation and task-mastery climate rather than competitive other-referenced environment [6]. Therefore, promoting collaboration and task-mastery—as opposed to competition—becomes an important design guideline in my family-centered technology-based intervention. More importantly, my design could be used to see how the omission of direct peer-competition would affect user's engagement in the context of physical activity intervention.

2.6.2 Physical Activity Reflections

At least two work have reported the presence of novelty effect [45,47], where the *gameful and playful* design elements no longer engage the users in the desired

physical activity level after the initial entertainment value starts to wear off. Therefore, to establish a more lasting impact on the PA intervention, I proposed to build health literacy using PA reflections while the technology's novelties have not ceased off. Previous work has provided reflection on PA, however these work relied to the user to independently undertake the reflection. In this research, I explore the design of a system that actively facilitates the PA reflection.

Collaborative reflection has the potential to model and influence healthy behavior in a family-centered intervention. Grimes et.al. [35] highlighted how collaborative health reflections allow parents to model positive behaviors to their children, and at the same time allow parents to receive motivating feedback from their children.

The value of reflection is also highlighted by Li [44]. On an automated PA logging system, where the responsibility to track PA data is taken over by technologies, regular reflection on physical activity pattern is potentially useful to maintain the awareness of physical activity opportunities. Grimes also highlighted that reflections affords the recall of rich contextual meaning of physical activity data, enriching the statistical numbers with the affective aspects of physical activity [35]. Grimes further elaborated that the rich contextual information helps families to attain the fullness of physical activity experience [35].

In the field of educational technology, reflection on concrete activities is the essential element of the experiential learning cycle, which was made prominent by Kolb [40]. As opposed to didactic learning where knowledge is told, in experiential learning framework learners are engaged to concrete activities, which are later reflected upon to scaffold new knowledge. Kolb described the four modes in experiential learning: concrete experience (CE), reflective observation (RO), abstract conceptualization (AC), and active experimentation (AE); and he described these relationship using Lewin's, Dewey's, and Piaget's model. Figure 3 shows the relationships mapped using Lewinian model.

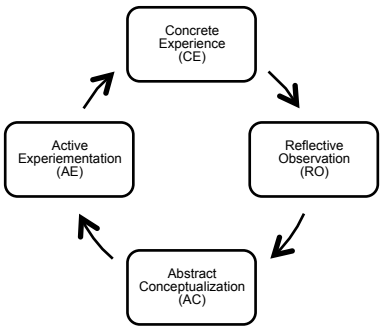

Figure 3 Experiential learning cycle based on Lewinian model

Previous work in HCI have clearly provided means for users to reflect back (RO) to past physical activity (CE), however these work have not attempted to actively direct PA reflection to think about the relationship of these CE's with it's

health benefits. The experiential learning framework and the availability of PA monitoring sensors motivated me to explore how the short window of novelty effect could be traded off with the user's attaining deeper understanding of physical activity intensity and duration.

Kolb argued that "the process of learning requires the resolution of conflicts" between current abstract knowledge (AC) and acquired concrete experience (CE), and between observations (RO) and actions (AE). He later emphasized that learning is a tension and conflict-filled process. The scope of this work did not allow the intervention to support all four modes of experiential learning. Instead, this research attempts to understand how concrete PA intensity data acquired by monitoring devices (CE) could be used to actively direct a focused reflection (RO) (as opposed to passive and undirected reflections in previous work) and let individuals to abstract their PA literacy (AC) by active experimentation (AE).

2.7 Summary

By drawing inspiration from these rich work in HCI, game design, health science, instructional learning, and SCT framework I concluded that a family-centered technology-based intervention has the potential to help families model and mutually-influence healthy behavior. Yet this has not been thoroughly studied by the HCI research community. In the context of exergame, obliterating the undesired effect of competition could be accomplished by promoting collaboration and design a task-centered competition. Providing physical activity reflection opportunity adds an additional value to expunge the "novelty effect", affords modelling and mutual-influence, and scaffold PA literacy.

Chapter 3
Identifying Design Guidelines

3.1 Introduction

Previous work on HCI and public health have motivated my research to use technology for promoting PA and guided my intervention design to use exergame in family setting. I recognized that obesity in United States is caused by numerous intertwined factors such financial barriers, environmental barriers, advertising, public policies, etc. [42]. Therefore, to design a technology-based intervention in this setting, I need to understand the attitude and the behavior of my target users toward physical activity. Several questions unfold and needs to be answered before designing the exergame. In this section, I described how my research partners and I conducted fieldwork research to form the basis of my design guidelines. This fieldwork also attempts to answer my research question 1: *What are the design opportunities for family-based technology interventions that promote physical activity?*

3.2 Formative Fieldwork Method

My research team conducted four focus groups sessions, which were followed by surveys. I assisted the interview session and recorded four pages of field notes following the sessions. We recruited 13 adult caregivers from the Family Gym (formerly known as Saturday Open Gym), a community program in Boston neighborhood that provides free and safe space for families to be active [3]. Family Gym is organized in a predominantly lower income neighborhood in Boston. My research partner conducted follow-up surveys, in which 10 people participated. Participants received $10 gift card for the group interview and an additional $10 gift card for the phone survey. The Northeastern University IRB reviewed and approved the IRB protocol.

3.2.1 Focus Group Interviews

In the focus group interview, participants were first asked to fill out demographic survey, and then we led them to participate in the interview session. There were two to six participants in each session. The focus group was aimed to gain insights to answer RQ1, such as: parental attitude and support toward kid's physical activity; experience with physical activity; and attitude towards technology.

The focus group interview data was recorded and transcribed, then my research partner and I conducted a qualitative analysis inspired by grounded theory analysis. We independently coded the transcripts and met regularly to discuss and reconcile our codes. We then iteratively clustered codes into higher-level themes and identified relationships between themes through axial coding.

3.2.2 Follow-up Survey

We administered a follow-up survey using validated instruments to gain more insight about parent's PA stage of change [82], child's activity level and support from parent [49], and neighborhood barriers to provide PA support [70]. The survey also asked participants about their access to technology. My research partners conducted the survey and provided descriptive statistics to lay out participants' attitude and perception towards PA and technology.

3.2.3 Participants Overview

13 people participated in the focus group. For the survey, 8 people completed the survey over the phone and 2 were self-administered by the participant. 3 people did not complete the survey.

All but one participant has access to the internet using either a home computer, tablet, and cell phone. Half of our participants reported to browse the internet using their phone at least once a day.

Table 3 describes the parental assessment of their child's PA. The parents reported that their child is generally active and has been active in the past 7 days. Table 4 details parental PA modelling and support. The parents indicated that almost every day they provided support and influence for their child to be physically active. However, lower value in "PA with kids" suggests that the parents are less frequently providing PA modelling. The parents reported high enjoyment of PA (average = 57.1, max possible = 63) and all parents were either in the action or maintenance stages of PA change. However, these self-report findings contradict their reported Body Mass Index (BMI), which mostly falls within overweight to obese range (n=11).

Number of days the child was physically active (n=10)	Average	Stdev	Min	Max
In the past 7 days	6.1	1.37	3	7
In a typical week	6.2	1.40	3	7

Table 3 Parental assessment of child's PA

How often the parent provided modelling or support for kids (n=10) scale 1-5; 1: never, 5: daily	Average	Stdev	Min	Max
Encouraged PA	4.8	0.42	4	5
PA with kids	3.7	1.10	2	5
Provided transport	4.5	0.85	3	5
Watched kids PA	4.1	0.87	3	5
Tell the value of PA to kids	4.9	0.32	4	5

Table 4 Parental PA Modelling and Support

Table 5 details neighborhood factors that affected child's PA. The parents reported that neighborhood safety is the major factor that keeps their child from

being physically active. Limited access to PA opportunities are also factors that parents felt undermined children's opportunity to be active.

Perception of neighborhood barrier (n=10) 1: strongly disagree, 4: strongly agree	**Average**	**Stdev**	**Min**	**Max**
Not enough space to be active in	2.2	1.40	1	4
No choice of activities	2.3	1.34	1	4
No equipment	2.2	1.23	1	4
No adult supervision	1.4	0.96	1	4
No other children there	2.0	1.25	1	4
Not safe because of crime (strangers, gangs, drugs)	2.7	1.16	1	4
Child gets bullied, teased, harassed	1.7	0.8	1	3
Not safe because it is close to a road	1.7	1.16	1	4
Too many people there	2.1	1.1	1	4
No good lighting	1.4	0.70	1	3
Difficult to get to	1.7	1.06	1	4

Table 5 Parental perceptions on neighborhood factors that affect child's PA

3.3 Results

Six larger themes emerge from the grounded theory analysis. The parents in our focus group seems to *recognize the value of PA* and this is the driving factor for them to take their child to FG. They provided *PA support* for their child and experienced a number of *PA barriers* on the process of doing so. These set of *PA barriers* are interconnected to each other and affecting the ability to be physically active. The *parents' emotional experience*s are elevated with their children when they are physically active together, suggesting the value of these experiences to encourage continuous PA engagement. In addition, parental *inclination towards technology* provides support for family-centered technology-based interventions.

3.3.1 Recognizing the Value of PA

It was evident that all 13 participants are aware of the health risk associated with obesity, and valued PA as a way to prevent obesity. Parents are concerned with obesity, such as expressed by P6, whereas other parents like P4 expressed how they intend to address the issue:

> P6: One main reason is that the boy, which is eight, he's a little overweight and I want to catch it early.
>
> P4: All of us, all of us are overweight because she [the grandparent] cooks three meals a day. So it's very important to her that he [the child] gets physical activity.

Most parent appreciate the physiological benefits of physical activity but a few of them recognize the psychological benefits as well:

> P5: I think it also allows them to concentrate a bit better after that physical activity is out of the way. The rest of the day they're a little bit more focused on whatever it is we're doing, whether it's cleaning, or we're giving them worksheets to do, they just seem more focused.

Parents are also making efforts to inculcate healthy eating habits. A few participants are also making a connection between instilling healthy behaviors in young children so that they can sustain it in the future as adults.

> P1: I feel like right now if we teach a child to eat healthy now and then they can eat like that later on.
>
> P6: So I want to get him started now and cut that out and hit him with the food, fruits and salads and so forth.

3.3.2 Parental Physical Activity Support

Most parents are concerned that their children's PA level is insufficient, and they feel the need to seek more opportunities than what are available through the school system. They also expressed concerns about insufficient PA during the weekdays, colder months, and the summer season.

Parents discussed in detail how they find appropriate activities during the week and look forward to spend time with their children (P7). This discussion also includes the various avenues that they have explored to find activities, like the YMCA and summer camps. One participant (P5) discussed that if she finds children playing soccer or happens to read a banner about physical activity events while driving around, she approaches them and gathers some information for her child.

> P7: We always go to places. I sometimes will go to the park, the beach, everywhere. I always spend time with them, any time I can I spend time with them.
>
> P5: I do feel I find some things, but I think when I'm driving around and I see someone playing soccer or something, I pull over and I ask more questions, I find out a little more, but I feel like there's information out there.

Accounts from four parents suggest that technology is one of the most explored options to find activities. I will discuss the role of technology in the following section.

The parents also expressed their concern about the change in environment and child's attitude, and how it affects their physical activity. One participant was concerned about her child's attention span and how that affects the type of activity that she prefers. Comparisons were made with the changing attitudes amongst children towards physical activity and that of their parents. For example, P5 said:

> P5: My parents, you know, walked me to the park, that was fun, but to them, that is boring.

At the same time, the parents discussed some of the challenges they faced in finding PA opportunities. These will be discussed in a later section. One of the participants brought up an interesting point where he emphasized that for him physical activity is not something that keeps the child engaged for some time.

3.3.3 Physical Activity Barriers

The parents reported to experienced barriers in providing PA opportunity for their children. One parent who worked for more than 40hrs/week, experienced *time*

barrier, the unavailability of time to provide physical activities for their kids. She views Family Gym (FG) as a good opportunity to be with the kids, to have fun and be physically active because FG is held during the weekend. One parent experienced *environmental barrier*, typically with regards to neighborhood safety. This discourages them to take the kids outside for physical activity. Our follow-up survey in Table 5 further suggests that neighborhood safety is the major factor that undermines child PA. Unfortunately, due to time constraints we were not able to probe this further. The limited affirmation of safety concerns in the focus group suggests that the parents may not be comfortable discussing the state of their neighborhood in social setting.

However the most apparent issues are *financial barrier* and *informational barrier*, where the parents experienced difficulties to find physical activities (such as summer camp and soccer club) that are within their budget limit. P7 expressed how difficult it was to find summer camps to engage her child in PA over summer.

> P7: And I just feel like it's a catch 22, because you want quality childcare, and you know there's cost associated with that, but sometimes it feels like things are out of reach.

P4 had a similar experience in finding activities, emphasizing that physical activities' advertisements are often misleading.

> P4: You read about them and then sometimes they don't even tell you there's a cost or what the cost is. Then you get there and surprise, it's $40, or surprise, you can do it free today but if you want to continue you have to sign up for it, and then it's $100 for the summer.

Sometimes *financial* and *informational barrier* are coupled by *transportation barrier*, requiring the parents a significant effort of transportation. This is further dampened by *time barrier*: not having sufficient time to take their kids for PA opportunities. These set of barriers parallels to the barriers identified by Sallis and Hovell [73].

It is evident that these barriers are not independently restricting their options, but are simultaneously interacting and affecting the parents' ability to provide PA opportunities for their kids. The interplay of all these factors results in the parents compromising on engaging their children in physical activity.

> P5: Some of the resources that are out there are either very far out and the drive is ridiculous just to get there, and I'm not driving to a situation where I come back to [a neighborhood in Boston] to go to work, or they're very, very expensive

Interestingly, two FG parents detailed the process of find physical activity programs on the Internet, suggesting they have developed the skill and mastery of curating hyperlink structure of the world wide web:

> P4: Yeah, it takes some research. You may find one activity, and then from that web page you find something else, and then from that web page

P4 comment about the tedious process of searching PA opportunity was also backed by P5. P5 state that her experience mirrored P4's experience:

> P5: You find something else, or you go to the activity and you find something

Accounts from the parents suggest that the barrier is not on lack of information, but rather to the abundance of it and how it could not meet their financial or transportation needs. This suggests that curation of PA information would be beneficial to the parents. This is further supported by the parents' keenness on seeking and exchanging information about potential opportunities with other FG parents. This desire of information is not limited to physical activity alone but also includes more generalized information about activities kids could go to or attend.

We explicitly asked the parents about the forms of support they received. However, it was not evident whether the majority of the parents received substantial support from their family or their community. Only one parent discussed receiving financial support from extended family members (P4) and two other said that they provided transportation for other parents.

> P4: My mom, she's 81 years old so, but she's still very active and she works, but I'm a student right now, I left my job so I could go back to school, and she makes sure that we keep my son involved in sports. Like she'll pay, he's going to camp all summer. It was $135 a week and she totally like paid for all of that and said it's very important

These suggest that most SOG Parents did not receive support or did not seek support from their family or community. They may also not comfortable discussing the support they received in a group setting. However, discussions during the focus groups suggest that they have been overcoming the barriers individually. For example, P4 and P5 are overcoming informational and financial barrier by developing their skills to curate information on the internet.

3.3.4 Emotional Experience during PA with Child

All of the parents had a *positive emotional experience* when they are physically active with their child. Their emotions are elevated when they exercised together and felt a positive stimulus from their children, which suggests the positive and mutual side effect attained during parental modelling. They also felt joy and satisfaction when they saw their child being active. Other parents saw PA as means to transfer and discuss family values (influence) as well as an opportunity for their children to make friends.

Many parents experienced a *one-on-one connection* with their child when they are physically active together and receive positive stimulus from their children. P4 expressed that she can be carefree and not have any inhibitions when active with her child. P2 who is visually impaired feels accepted when her daughter wants to be active together.

> P7: I love to spend time [exercising] with my kids and I can feel that they are really feeling it too when they're playing with me.

> P4: I also feel silly [when exercising with my child] because we're having a good time. We're real goofy together.

> P2: [I feel] proud because my daughter is not ashamed of me. I can't see [..] and do things together, and it doesn't bother her to have me there.

The parents also feel connected when they are observing their children engage in physical activities. The *observatory satisfaction* is heightened when they see their children develop physically and hone their skills over time.

> P6: I remember when he was shooting the basketball he couldn't even come close to the net. Now he can, and he gets excited and I'm like okay, he's developing. He's maturing and the rush I get out of it. I get excited being around him.

> P4: I feel proud because he likes to swim and he took to it from the time he was an infant in the bathtub, and we took him to swim lessons, so he's a great swimmer, so I feel proud to see him active.

Other three parents saw physical activity as an opportunity to *transfer values and skills* to prepare their kids for adulthood. For example P3 and P8 expressed the satisfaction of being able to transfer values through physical activities:

> P8: By doing this [physical activity together] and consistently doing it you build a good foundation to be strong so at the end when my daughter be grown. I know she's gonna be a strong individual.

> P3: I feel like I'm teaching him something. I'm learning what he enjoys to do, what he likes and doesn't like. Also I feel like I'm coaching him, coaching him the right way, the proper way of doing things.

> P11: As a parent we need to help be the first teacher, the first guide to having those healthy habits.

They all agreed that they experienced togetherness and were more connected with their child, which in turn improved their interpersonal relationship. They were happy and proud to watch their children, be physically active. This quote aptly summarizes the experiences the parents discussed during PA:

> P11: I think that there's a general sense of joy knowing that you're doing something good for your family and you're providing in a healthy productive way. And so when we are together, the four words, I said healthy, happy, intentional.

3.3.5 Inclination Towards Technology

We also explored how FG parents perceive technological gadgets and Internet. Our finding suggests that FG parents are accepting their children's strong affinity with technology and are aware that limiting technology at home would be ineffective because their kids would have access to technology from their schoolmates anyway. Therefore, the parents are incorporating technology into their households by setting the rules for what kind of technology that can be used, and when the kids can use it. They discussed in detail about the type of technology they prefer and make comparisons with other forms. Their choices are heavily influenced by their perception of the benefits that their children may have through some educational applications. Hence the use of technology is encouraged when the parents foresee a certain degree of benefit. They discourage the use of violent video games, excessive TV viewing and non-interactive video games.

> P8: They [the children] are not gonna do anything that they're not supposed to because if I catch them on anything they're not supposed to they lose their computer.

> P10: She has half an hour watching TV because I don't believe watching TV is the kind of thing she should be watching a lot. So after she come home she will do her homework and she go to the computer and learn how to read and we have those things that she can jump like I don't know how you call it.

Several parents incorporate technology further by using gadgets and educational video games to assist in parenting. Several parents are using gadgets as a "baby-sitter" to substitute their presence so they can do other work. Other parent is giving access technology to her child to reward desired behavior.

> P9: It is kind of like "baby-sitter" whenever I'm in the store and my son is spazzing out I give him my phone.

> P11: I just need to take a shower and you all need to watch an episode of Fresh Beat Band because they're always singing and dancing on that show.

> P12: I'm beginning to use it as like a bargaining tool, like listen you do not use my phone if you don't do this, if you're not good you know what I mean?

However, P12 who used gadgets as a bargaining tool expressed a conflict of values over her discretionary use of technology. P12 felt that she is not supposed to give her child access to her smartphone as a bargaining tool and at the same time every other kids plays video games. She expressed her conflict of values in disciplining her child:

> P12: So I'm kind of I feel a little guilty about it because I don't really like it but every kid in the country in the world practically is into it. So I'm a little torn.

3.3.6 Social Networking Sites as a Potential Source of Support

The Family Gym organizer set up a Facebook page on November 15th, 2012. However, a majority of the parents were not aware its existence. Most of them heard about it for the first time during the focus group session that we conducted in June-July 2013. The parents who knew about the page have not accessed it or use it at all. When we probed further, they expressed their willingness to use it actively in the future and post information. For example, P12 said:

> P12: That [Zumba class] is another thing I could share on the Facebook too. At least around where I live—I live in [a name of a neighborhood in Boston] —and they have free yoga, sometimes it's Northeastern stuff, sometimes it's just BHA.

However, they seem to look at it as an administrative portal to share information. They expect some level of administrative initiative on furthering the use of the Facebook page. P8 suggests the FG to promote the Facebook page using the bulletin board:

> P8: Maybe you ought to put something on the bulletin board to let them know you can go to dot whatever [FG Facebook address] just because the parents are always flipping and staff might forget. Put it up there or put it with the stuff that we give. Don't forget to go on Facebook or whatever.

This is an important clue of how FG parents perceive such type of social networking platform.

3.4 Design Opportunities

From the interview analysis, I summarized the design opportunities when designing a technology-based intervention to help families model and mutually-influence healthy behavior.

3.4.1 Attitude Towards Technology

The parents provided insights how technology are integrated into their home. The parents felt that their child preferred indoor activity. They described their child's affinity with technology and they how set the rules how technology like computers, videogames, smartphones, and tablets could be used by their children. They also controlled what kind of content that can be accessed, and this decision seems of be determined by perceived benefits of that technology. Some parents even use technology as a bargaining tool to control their child's behavior and at some extent as a "baby-sitter" to keep their child busy so the parents could do other errands.

These findings provided further support for the use of technology for promoting PA in family setting. This also provides support that parents need to be explicitly informed about the health benefits of exergames. While physical activities could be seen as an indirect benefit of sport games, in the context of exergame parents may moderate what kind of technology that their child can use thus raising the need for explicit deliberation of benefits.

3.4.2 Technology as a Platform for Modelling and Influence

Our interview and survey data indicated that the parents provided support for their child to be physically active by actively seeking PA opportunities and providing transportation to PA venues. They also influenced their child to be active and sometimes used PA as an opportunity to influence family values. However, the interviews suggest that the parents may not be actively providing PA role modelling, and this is affirmed by the survey data on parental modelling. We looked into an internal study conducted by the FG organizer in 2010 and found that most FG parents did not provide PA modeling during the sessions. Using SOFIT [48] to measure PA level, the study found that most of the time parents are either sitting or not engaged in any activity. This is further supported by evidence from the survey data that most of the parents' BMIs are within the overweight and obese range. This raises concerns that the parents may not engage in recommended PA at all.

From this finding, I inferred that while most of the parents provided PA support for their child by seeking PA opportunities and providing transport to PA events, they might not provide PA modelling. Using SCT framework, this could be described that parents provided facilitation for PA but did not provide observational

learning opportunities for their child. Many factors could explain why the parents were not physically active with or without their child. Study by Brownson [12] found that the top PA barriers for lower-income adults are: perceived had enough PA at work, too tired, and no time. Going back to the recommendations proposed by Thompson to accommodate the complex needs and demands of today's families [89], a technology based interventions would allow parents to reflect to their PA level with their children (even though they are not active at the same place and time) and use that reflection process as a PA modelling opportunity.

We also found that the parents were pleased when they are physically active with their child and actively providing PA support for their child. This opens an opportunity to design a social exergame, in which the goal could only be accomplished if both the child and the parent are physically active. Since influence at home works both ways [35,94], providing a collaborative PA reflections and collaborative exergame could potentially add motivations for the parents to be physically active.

3.4.3 Technology as a Platform of Interfamily Social Support

The FG Parents expressed their desire for more social interactions with other families. However, interfamily interactions could potentially induces privacy issues and in exergame context may lead to the undesired effect of competition, especially for games that is deployed in public setting.

3.4.4 Concrete Activity Data for Active Scaffolding

While the interviews, survey, SOFIT analysis by FG organizers, and BMI data gives indication that the parents are not physically active, this is contradicted by the parents' reported stage of change. Using a validated survey that measures physical activity stage of change [ref], we found that the parents self-reported as either in action or maintenance stage. This discrepancy may suggests the presence of a gap between perceived PA level and the actual PA.

Previous work have explored the use of activity tracking device and novel PA visualizations, and how it raises the awareness of physical activity level [18,45]. However, as discussed in previous chapter, these visualizations are not taking an active role in informing the relationship between PA and its health benefits. A potential design opportunity is to use the concrete activity data to aid scaffolding this relationship. The cycle of experiencing concrete activity and reflection on that activity affords experiential learning cycle [40].

3.4.5 Emotional Aspects of Physical Activity

Our participants provided a rich account on their emotional experience when they are physically active with their child. The unifying theme is connectedness between parent and child, however other accounts also provide intimate emotional experiences such as proud and acceptance. Parents are also using PA moments as an

opportunity to transfer values and knowledge to their offspring. This is a preliminary evidence that physical activity should be seen beyond the conventional ways of calories and steps; and expressed as emotional moments which could be materialized as artifacts to be reflected back upon.

3.5 Summary

The result from this formative study gives insights for me to design an exergame for promoting physical activity in family setting, solidified the potential of collaborative reflections, and provides a partial answer to my RQ1:

Automated tracking devices allow people to automatically log their PA level and see their activity graph and charts. This data could potentially be used by parents to provide PA modelling even though they could not be physically active at the same time and place with their child, especially for modern families who are facing the challenges to physically active with their child at the same time. Having a collaborative exergame for parent and child could potentially promotes the child as a source of PA motivation for the parent. The activity data, composed of concrete PA artifacts, is potentially useful to aid scaffolding the relationship between PA and its effect on health. This is particularly useful to narrow the gap between perceived and actual PA. Additionally, PA data should be viewed in a larger and richer context as moments, in which Grimes said "to attain the fullness of physical activity experience" [35].

Chapter 4
Spaceship Launch: Design and Evaluation

Based on the formative fieldwork I described in the preceding section, I decided to design Spaceship Launch, a collaborative family-centered exergame for promoting physical activity by leveraging the mutual influence between parent and child. The exergame also attempted to overcome the "novelty effect" problem and "gap of perceived PA" problem by using PA reflections to actively scaffold the value of PA before the game appeal wears off.

In this section I will describe the game mechanics and design rationale that shaped my system based on the insights from previous work and my preliminary study. Then, I continue with describing the system in more detail using user interaction scenario and technical descriptions. After that I will provide the methods to deploy and evaluate the system, and finally present the result of the evaluation.

4.1 System Description

Spaceship Launch (SL) consists of two elements: the Dashboard and the Launch game. The purpose of the Dashboard is to provide collaborative PA reflections between parent and child, whereas the Launch game provided the virtual rewards and articulated the game's task in which the team of child and the parent must accomplish. Figure 4 shows schematic design of the SL Dashboard.

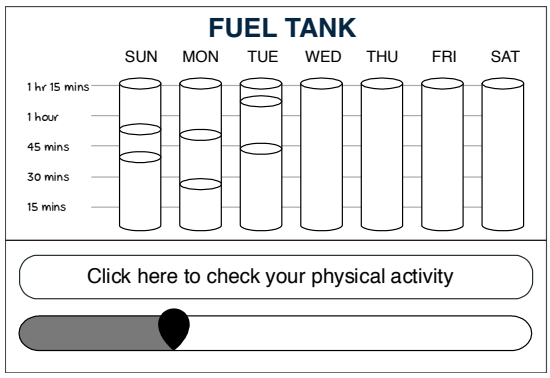

Figure 4 Schematic of Spaceship Launch Dashboard

The task of this game is defined as follow:

> Collect as much spaceship fuel as possible to launch your spaceship to planets in outer space. You can get enough fuel to launch to one planet if you and your child are physically active in a duration that meets CDC's recommendation per week.[1]

The parents and children participating in the game were asked to wear Fitbit Flex and Fitbit Zip, respectively, to acquire their physical activity duration and intensity.

4.1.1 Game Mechanics

The game is designed to be *collaborative* for a two-member team. Both players can only accomplish the goal (winning) or not accomplish the goal (losing). With this mechanic, there is no instance where one player wins and the other lose. Using SDT framework, parent-child team is selected so that the feeling of *relatedness* would improve the motivation and enjoyment in using the game.

SL is using a reward system as the base game design element in Deterding's classification. The reward is obtained if the both players engaged in PA that meets CDC's recommended duration and intensity. The form of the reward is the ability to launch the spaceship to a planet and the experience of *fantasy* and *discovery* of the planet's inhabitant and environments. The planets and overall game visuals are designed towards the kids, because they are expected to be the driving factor of engagement. The reward system is non-punitive, or in PLEX framework's term: *non-cruel*. The spaceship will not be sad or damaged if the player is 'losing'. This decision is informed by research on SDT where negative feedback may have disadvantageous impact on competence, which is evident in Fish'N'Steps where the player become disengaged when seeing a sad fish.

The design decision to make the game collaborative also omitted the presence of leaderboard game design element. To reiterate, accomplishment on this game is driven by task-mastery (or task-completion) rather than peer-comparison. This is to avoid the undesired effect of competition.

The higher level game design element is the time-constraint game-mechanics: the players are limited to launch their spaceship at the weekend. Therefore the players have to meet the PA requirement in 7 days; otherwise they will not be able to obtain the reward.

Upon landing at one planet, the planet inhabitants greets the children and then asking for help to complete a mission. This mission is a simple steps game: the child must be active to collect steps to get the planet inhabitant's favorite food. In this study, the target was set to 800 steps, based on the ideal 9,000 steps per day for

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[1] For this study I am using CDC's recommendation of 150 minutes per week of

adolescent [2]. The number of steps is taken from the Fitbit. This mission serves the purpose of *closure*, giving a sense of completion for the entire spaceship launching experience. Steps based game is selected to disengage the child from the video game and encourage them to engage back to physical activities, fulfilling the purpose of SL to promote physical activity.

4.1.2 Design Rationale

The SL Dashboard is designed for the parent and child to monitor their physical activity levels of the week and their fuel progress. The rationale for this design is some families may not be able to be physically active together, even though the parent may be physically active at work and the child may be active at school. Having a technology as a venue to reflect and observe physical activities could potentially stimulate mutual PA influence for the family. Having the child to observe their parent's PA level could also potentially motivate parents to model positive PA behavior to their child. *Observational learning* is a central element of SCT [33]. Furthermore, the PA visualization is designed for *self-monitoring,* which affords *self-regulation*, which is also an important element of SCT. Our preliminary study suggest that the parents were experiencing elevated emotion when observing their child being active, therefore the dashboard may provide an additional incentive for the parent.

The SL Dashboard also prompts the parent with trivia questions to reflect back to a physical activity episode of that particular day. Then the trivia will ask them to guess how many portion of food it burned off. The purpose of this design is to actively scaffold the relationship between concrete PA experience and food intake. The cycle of experiencing concrete activities and reflections on the experiences affords experiential learning cycle, which was made into prominence by Kolb [40]. Figure 3 visualizes the experiential learning cycle. The physical activity episodes experienced by the parent are the concrete experiences (CE) whereas the reflection in SL Dashboard is reflective observation (RO). While previous work on has provided this reflection opportunity, none have attempted to actively scaffold the relationships between physical activity and the health outcomes. The PA perception gap found in in our preliminary study also provided an additional support for active scaffolding.

To promote reflective observation, the trivia question is phrased as follow:

> Think about your physical activity today at 9.11am. You did 16 minutes of <u>moderate physical activity</u>. How many Big Mac does it burn off?
> - 1/10
> - 1/3
> - Half
> - One

User can choose one of the four options. When they selected the right or wrong answer, the game will present some additional information to scaffold the concept of calories and its' relationship with food and physical activity:

16 minutes of moderate physical activity = 1/3 Big Mac. Your 16 minutes of moderate physical activity, equals to 165 kcals. One Big Mac contains 550 kcals.

The second element of SL, the Launch game articulated the game's task in which the team of child and the parent must accomplish. In the Launch game the players could see how many planets they can launch to, select the planet to launch, observe the launching experience, landed to the planet, and complete a mission.

Our preliminary study found that the parents accessed the internet using many kind of devices like computers, tablets, and smartphones. The preliminary-study also suggests that some parent of our demography may not have access to internet at all. To facilitate these diverse needs, I chose to implement the exergame such that it was accessible in a multi-modal environment: desktop computers, smartphones, tablets, and public-access large-screen display.

4.1.3 User Interaction

Every day, the parent and child could log into SL Dashboard (Figure 5) to see how many minutes they spend in physical activity. The daily minutes in PA is visualized as seven flasks containing the fuel acquired on each day of the week. The fuel acquired by the parent and the child has distinct color so they could see how much they contributed to get the fuel. Underneath the flasks is the fuel tank meter that visualizes the total fuel acquired over time and whether they have enough fuel to launch one or more planet.

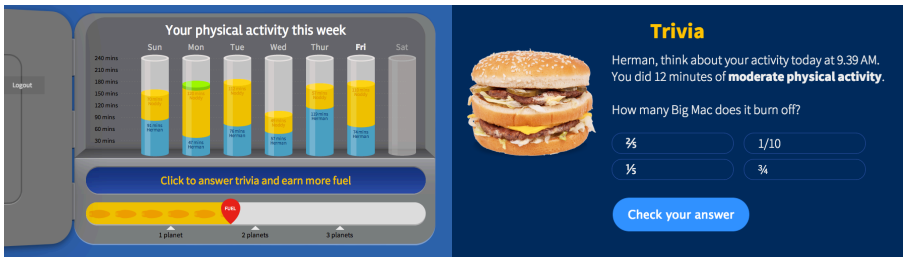

Figure 5 Spaceship Launch dashboard and trivia

During the weekend, the players could open the SL Launch game (Figure 6) in which they could enter the cockpit and see how many planets they can launch to. Upon selecting the destination planet, the child could tap on the "Launch" button to launch their spaceship. Then, they will be presented with an animation of their spaceship flew in outer space and landed on the chosen planet. When the spaceship landed to the planet, the creature will greet the player. The creature also asks the player to find their favorite "mystery" food. The mission could be completed by being active to collect certain amount of steps. Upon completing the mission, the creature will reveal its' favorite food and offer the kids to send the recipe to their parent's email.

Figure 6 Spaceship Launch game

Participants could access the Dashboard from desktop computer, smartphone, or tablets. The Launch game could be accessed from their home using desktop computer, smartphone, or tablet; and additionally at the Family Gym (FG) session every Saturday using large screen display. The FG is organized to provide safe, accessible, and free space for parents and children to be physically active [3].

4.1.4 Technical Implementation

Spaceship Launch was implemented on web-based platform and it consists of two components: the frontend and the backend. The frontend was developed using HTML and SVG, which is controlled using JQuery and SVG.JS JavaScript libraries. The frontend sends and retrieves data using asynchronous calls to the backend. The backend was developed using Laravel PHP Framework. The purpose of the backend is to serve the frontend, store the data, and retrieve user's PA data from Fitbit's server using OAuth protocol. The data retrieved from Fitbit is stored using MySQL database. We acquired a special authorization from Fitbit to access the minute-by-minute time series data.

The Fitbit tracks physical activity movements and estimated the steps, calorie expenditures, and minutes in physical activity. The Fitbits will automatically send the data to a wireless receiver, which will send the data to the Fitbit server. Fitbit accuracy has been validated by a number of studies [38,84].

4.2 Method

I evaluated the use and impact of Spaceship Launch in a Boston area neighborhood, in a 4-week field study of 13 families. These families were recruited from the Family Gym and is comprised of 29 participants: 15 caregivers and 14 children. Northeastern University's Human Subject Research Protection Office reviewed and approved the IRB protocol.

At the beginning of the study participants were asked to wear the Fitbits, showed the SL game and provided a short overview on how to use the dashboard. Then I asked them to fill out a validated survey to assess their PA intention [19], PA stage of change [50], parental PA modelling and support [49], and food healthfulness literacy [13]. I also asked them to provide demographic data. In total, the parents and kids participated in the study for 4 weeks. I asked them to wear the Fitbit for 1 week as a baseline. Then at the beginning of the second week I gave them

access to the game for 3 weeks. Their activities using SL Dashboard and SL Launch game are automatically logged by the system. While I asked them to log into the game at least once every week, I recommend them to check the dashboard as often as possible. Every three days I send emails or SMS reminders to the partcipants to log in if they have not logged into SL for more than three days. At the end of week four, I offered the opportunity to continue use the SL and wearing the Fitbits. Seven parents opted to continue using SL and the Fitbits. During the Family gym sessions I collected 1 page field notes.

Figure 7 Adults and a child using Spaceship Launch

Starting from week three, I interviewed eight parents to gain insights about their experience with the game and how it affected their PA intention and PA modelling behavior. The interview was semi-structured. I followed the interview with participatory design workshops with five parents and four children. In this workshop they were instructed to redesign the SL Dashboard using paper and craft materials so that the game fits better to their needs. Participants of this workshop ranging from one to two families, each had their children with them. The structure of this workshop is partially inspired by the work of Yoo [98] to elicit nuanced requirements and stories otherwise unattainable in conventional interviews. In this participatory design workshop I initially presented the parents and the children with an instruction card to guide them design their own SL dashboard. The purpose of the instruction card is to give a focused direction about what to design. It contains the following instruction:

> "Redesign the Spaceship Launch's dashboard so that you and your child could sit together and monitor your weekly physical activities"

After they completed the first task, I presented them with an envisioning card to help them revise their SL dashboard design. The purpose of envisioning card is to stimulate focused ideas to improve the existing design.

> "Improve the way your dashboard works so that it fits better with your routine, your household, and your family's needs"

After the families completed their revision, I conducted interviews with the participants. I used their designs to probe deeper and gain more understanding how

the game affected their behavior and how could be improved to accomplish its purpose.

The interview data was transcribed and then I conducted a qualitative analysis inspired by grounded theory. First, I inductively coded the participant's responses that result in a set of emerging themes. After that I conducted axial coding to examine the relationship between the themes. I also computed descriptive statistics for the survey and system log data. The result of this analysis is presented in section 4.4.

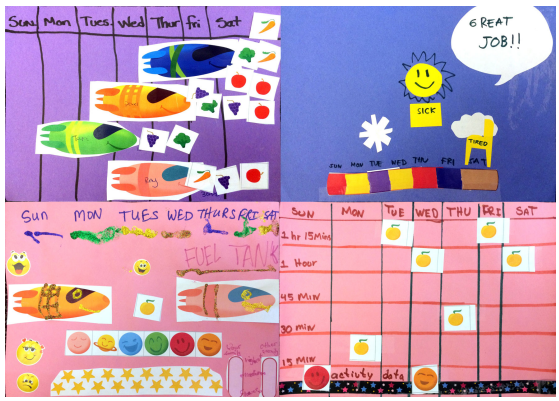

Figure 8 Design results from participatory design workshop

4.2.1 Participant Overview

I recruited 15 caregivers and 14 children from 12 families through four Family Gym sessions. The majority of the caregivers were female (n=11) and the majority of the children were female (n=10). The median age of the caregivers was 39 and the median age of the children was 8. The BMI calculation indicated that 8 caregivers are obese, 3 are overweight, 2 are in the ideal range, and 2 provided no data. The median household income was $20,000-$29,999 and the majority of the families were living in a predominantly lower income Boston neighborhood (n=13).

Parental intention to be physically activity appears to be low. Our survey data indicated that the median physical activity intention is a total of 5-8 physical activities in the next 4 weeks. However this is contradicted with the stage of change data which suggests the majority of the caregivers are in either action (n=5) or maintenance stage (n=5). Other 3 participants reported to be in the preparation stage, and 2 are in contemplative stage. This finding matches our earlier finding about the presence of a PA perception gap among our participants and this is further supported by the BMI data.

Table 6 described parental PA modelling and support for their kids. The data mirrors our preliminary study where the parents provided PA support for their kids (in the form of transportation, encouragement, and influence) but do not provide much PA modeling by participating in physical activity with the kids.

How often the parent provided modelling or support for kids (1-5; 1: never, 5: everyday)	Average	Stdev	Min	Max
Encouraged PA	3.40	1.06	1	5
Participate PA with kids	2.80	1.15	1	5
Provided transport	3.33	1.18	1	5
Watched kids PA	3.07	1.03	1	5
Tell the value of PA to kids	4.00	1.07	1	5
Tell about parental PA to kids	3.20	1.21	1	5

Table 6 Parental PA Modelling and Support

Two families do not have computer and internet access at home. As a result one of these family decided to opt out from the study. Technology literacy seems to be an major issue as 7 families had problems setting up the software driver for the Fitbits.

4.2.2 Accessing Spaceship Launch

Table 7 detailed how the families accessed the SL Dashboard and SL Launch game: more families accessed SL using public-access large screen display rather than using home devices (computer, smartphone, tablet) at home, suggesting the merit of providing multi-modal access for this demographic.

Access using home devices	Access using public-access screen	Both	No access
6	10	3	3

Table 7 Families' access to SL

While SL is designed as a casual game with minimal set up, to be able to monitor SL Dashboard daily, users need to install the Fitbit software. Table 8 detailed the instances of issues that were experienced by the families when accessing the game at home. The most common issue is installing the Fitbit software. I always offered the opportunity to assist home installation, but only 2 families accepted the offer. After families solved the Fitbit installation, they often disconnected their Fitbit from the game by setting up own private Fitbit account. This results in the account previously linked to the game to become disconnected. Two families do not have any computer and two have specific issues (non-compliance and unwillingness to install third party software).

Cannot install Fitbit	Disconnected Fitbit from game	No computer	Other
7	3	2	2

Table 8 Number of families who experienced access issues at home

Table 9 overviews our system usage log analysis. During the 4 weeks field study, participants logged into the system 10.06 times in average. Eight participants

logged in more frequently than the median of 9. Only 4 people accessed the trivia more than once. In average, the families launched the spaceship 1.31 times and 4 families were not able to launch the spaceship at all.

	Logging into Dashboard	Viewing Dashboard	Answering Trivia	Launching Spaceship
Average	10.06	9.56	1.19	1.31
Stdev	9.90	13.84	2.01	1.25
Median	9.00	3.50	1.00	1.00
Min	9.00	0.00	0.00	0.00
Max	29.00	53.00	8.00	4.00

Table 9 Overview of System Access

Deploying the public access screen with children as the target user is particularly challenging. There are numerous instances when several kids flocked in front of the screen and simultaneously tapped into the game's touchscreen. As a result, the game advances without the control of the child who is currently playing and the kid did not get the complete game experience. Entering the password using the large screen is also problematic. There is one instance of bullying when an older kid managed to peek into a younger kid's password. Then the older kid bullied the younger kid by logging in into the younger kid's account.

4.3 Results

Four higher-level themes emerge from the interview and participatory design workshop data. It is evident that for some parents, the game promoted *Increased PA intention* and the learning experiences within the game *Scaffolded PA literacy* for some users. A majority of the parents expressed their desired for more competition in this game, even though the game is designed to be non-competitive. As I explored further, I saw the *Nuanced Preferences of Competition* by looking closer into the multifaceted nature of it. Eventually it is evident that by exploring the various *Psychological Needs in Exergames* I was able to identify caregivers' desired elements in an exergame. It appears that competition is an integral element of exergame in family context and the data suggests competition design that produces positive outcome.

4.3.1 Increased PA Intention

The SL Dashboard provided an opportunity to keep track of daily PA and to check whether the PA level meets the weekly goal. Wearing the Fitbit and gaining PA visualization from the SL Dashboard seems to increase the *intention to be active*. However, these intentions are different from one person to another. Some participants (P01, P02, P03) reported the intention to exercise at individual level, whereas for a few others (P09, P01) the intention is at a team level (i.e. exercising simultaneously with the child as a team).

P01: Yeah, it makes me want to do more. Yes.

P02: Having it has made me want to, you know, be more active

P03: I'm thinking to exercise more. Because my fuels were low. And I know I could do, I know I could do better than that.

P08: [It made me say to myself] "Oh, I got to get up. Got to get out of the house. Got to do something."

P09: You know, you get your mindset. "*Okay, tomorrow I'm going to fill this tank up*" […] My daughter and I, we said "*Okay, we're gonna work*."

P01: Then I'm trying to think of things that we can do together. Um, to make, you know, more steps.

Some participants translated these desire into *actions to accomplish game's task*. The participants realized that the Fitbit tracks all kind of physical movement; therefore they come up with a very diverse range of exercise opportunities. To gain more fuels in the Spaceship Launch, P01 took the stairs to go from the ground floor to the 7th, P02 did Zumba and shoveled snows, whereas P16 walked instead of taking the bus.

P01: I made it a point to go to the seventh floor to get water. So, three times a day I go to the seventh floor. It makes me drink more water. And I feel like it's for a cause to go.

P02: Oh, this is a good opportunity for me to get some, *you know*, points for my [spaceship]." So it helped to encourage me to want to go do Zumba and then with all the snow was there I was like, "*Oh, I'm shoveling*."

P16: I just walk instead of catching the bus for a short ride. "*I'm gonna walk*." So walk there, and walk back.

In addition, to get more fuels P01 take her kid for an exercise together.

P01: So I got this tape, Shaun T. And it's a hip-hop moving tape. And I decided in the morning that's what we're gonna do. So I did it, and it worked one day.

During interviews, only one participant, P04, stated that the game did not affect her intention to be active. P04 expressed her appreciation towards the novelty of the game and found some use of it, but she indicated that the game did not elevate her motivation at all. She believes to be an internally motivated person, so a game would not affect that.

P04: I think when I exercise I kind of go in spurts, and either I'm really into exercising and exercise like four times a week, or I just don't exercise. And right now I'm in a phase where I'm just lazy, and not exercising.

Using the same pre-intervention survey, I collected interview participants' PA intention and stage of change (Table 10). PA intention is the total number of days in PA in the next 4 weeks. One participant did not carry out the post survey.

Participant Id	Pre-study PA Intention*	Pre-study Stage of Change	Post study PA Intention*	Post-study Stage of Change
1	21+	Action	21+	Action
2	1-4	Contemplative	1-4	Contemplative

3	5-8	Preparation	-	-
4	1-4	Contemplative	1-4	Contemplative
6	5-8	Action	9-12	Preparation
8	9-12	Action	21+	Maintenance
9	1-4	Action	5-8	Action
16	5-8	Maintenance	21+	Action

Table 10 Interview participant PA information

All participants who filled the post survey reported the same or higher PA Intention. This suggests that the combination of participating in the game and wearing the Fitbit have some effect in increasing PA intention.

4.3.2 Scaffolding PA Literacy

The trivia component in SL Dashboard was designed to actively scaffold PA literacy. However the Dashboard on its own affords reflection and produces various PA literacy outcomes. At the basic level, SL Dashboard encourages participants to *remember* their physical activity episodes. P02 reported that when she was looking at the chart in SL Dashboard, she was stimulated to reflect back to her activities and remember what kind of activity that causes the number to go up or down. She also looked deeper into the PA data by remembering the places she went to as well as the frequency of her PA.

> P02: When I look at whatever the dashboard says it causes me to think, "*Okay, what did I do today?*" Or, "*Where did I go today?*" Or, you know, "*Was I sitting around more today or was I up and down more today?*"

P03 also described a similar experience. The chart in the SL dashboard helped her to reflect back and think about the activity that causes the number to go up. And when the number is low, she described her aspiration to be more active.

> P03: I liked to see the activity [monitor], because it helped me see, you know, what I'm doing and what I can do more of.

In the SL Dashboard, participant only saw the minutes in moderate or vigorous PA at daily interval. They do not have access to hour-by-hour or minutes-by-minutes PA data. With this limited overview, P02 was stimulated to think back the specific activities that cause the numbers in the dashboard to fluctuate. P03 also implied that the SL Dashboard helped her to see how active she was.

The day-by-day visualization affected a few participants to make *comparison* from one day to another. P03 described how she was less active in one day and compare that with a more active day.

> P03: [When I saw the game dashboard] some days I wasn't active and some days I was more active.

P02 however, demonstrated a more detailed understanding by looking deeper into activities (rather than just looking at days) that affected the minutes in one day to go higher compared to another day.

> P02: it gets me to think, "*Okay, what did I do today?*" That may have, you know, caused one day to be higher and another day to be higher? […] I remember the day that did, the last time I did do Zumba, that was an activity. That was a day where it did show a lot more activity wise.

To categorize the emerging learning experiences I borrowed Bloom's revised taxonomy of learning outcomes[2] [41]. Bloom taxonomy's is a set of learning objectives to aid curriculum and examination design [11]. A revised version by Krathwol [41] is used in this work. These learning objectives are: remembering, understanding, applying, analyzing, evaluating, and creating. The ordering suggests a hierarchy that one objective is more complex than another (i.e. remembering is less complex than understanding). However, Krathwol suggests that the distinction is relaxed and one category may overlap with another.

It appears that two outcomes emerge when the participants were using the SL Dashboard. The SL Dashboard aided the participants in *remembering* their past PA. In addition, the participants also experienced a gain of *understanding* of PA by making comparisons of one day and another. However, when using the active scaffolding in the trivia, the SL Dashboard seems to afford deeper understanding of PA literacy. It appears that a few participants were *analyzing* the relationship between food and activity, and formed a new knowledge. For example, P09 was surprised that her 15 minutes of light PA only burned off a half banana. Then she started to show an indication of *organizing* PA by intensity type and the food equivalency.

> P09: I would think like the 15-minute light activity would [burn] a whole banana. I mean, after all it's a banana, you know. And so I was like, those kinds of things like shocking, you know. Between moderate and light, it makes a lot of different I guess, you know.

Similarly, P16 was able to express the association between walking and Big Mac even if the relationship was still inaccurate because 16 minutes of moderate PA only equates to 1/3 of a BigMac. This could be attributed to the way the scaffolding information was phrased or laid out did not fit to his cognition.

> P16: I learned if I eat a whole Big Mac, I would have to walk about 16 to 20 minutes to burn all them calories off I just ate, I've eaten.
>
> I: Does that surprise you or you kind of know about this before or …?
>
> P16: I didn't know about it. I learned it there.

The various types of learning experiences that emerge from using the game with or without active scaffolding suggests the opportunity of experiential learning using concrete PA data.

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[2] Bloom's revised taxonomy of learning outcomes: remembering, understanding, applying, analyzing, evaluating, creating

Participant's account also suggests the emergence of conflicts between old PA and diet knowledge and new knowledge. P09 expressed a surprised reaction when realizing that 15 minutes of light PA only equates to half banana. Indeed, learning is also the process of unlearning, which is described by Kolb as tensions between concrete experience versus abstract conceptualization and active experimentation versus reflective observation [40]. In P09's case, her abstract understanding of PA equivalence to a banana seem to be less that what she discovered from reflection of the concrete PA experience. These dialectic tensions may also occur between active experimentation and reflective observation. Kolb further argued that the way these dialectic tensions are handled affect the learning outcomes. P09 appeared to reconcile this tension by acknowledging the difference between moderate and light PA:

> P09: Between moderate and light, it makes a lot of difference I guess.

4.3.3 Nuanced Preference of Competition

Spaceship Launch was designed with no prominent in-game competitive element. However, the interview data suggests that competition is a desired feature. Five participants proposed competition element to the game even if I did not ask them specifically if they wanted a competition. Many participants saw the value of competition to make the game fun and to add more motivation to play the game.

> P01: I think competition will make it more fun.

> P03: Yeah, more competition type of thing, cause sometime when you see that someone else is doing better, you want to do better, more better.

> P04: I think it might motivate me to have like a little competition with her [the child] of who was more physically active.

> P06: You know, and *uh*, you can make a competition if you want.

> P09: I think competitions good. With each other. As long as they're competing for positive things.

It becomes clear that competition is multifaceted and the definition could mean many things. While previous work in HCI and health behavior warned the negative outcome of competition, some form of competition may improve exergame's engagement. In this section, I will describe the various kind of competition that emerged from my data, and how specific form of competition may actually beneficial. In Figure 9, I categorized the emerging competition types by the setting in which the competition occurred: intrafamily and family-to-family.

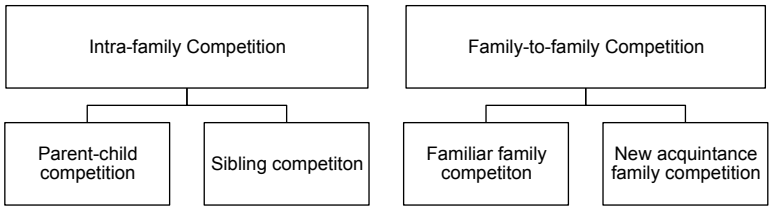

Figure 9 Competition types

Intrafamily Parent-Child Competition

The most discussed form of competition is *parent-child competition*, suggesting the higher need of it. P01 appeared to devise a competition with the child when the game did not provide the option to do so:

> P01: I'll go, "Look, you won, and you made, you did more activity then I. You made the, you get to the planet more quicker than I did, because you did XYZ."
>
> And he'll go, "Oh, you need to exercise more mommy."
>
> And I'll go, "*Yes, we do*."
>
> And, you know, he'll go, "You can't beat me, I'm a child."

In the conversation above, P01 showed the SL Dashboard to her child, and described how her kid outperformed her in in terms of activeness. That sparked a health-centered conversation that goes back and forth between the parent and child, thus suggests the value of competition to stimulate health-related interactions. Furthermore, P01 imagined how a parent-child competition could be afforded by the SL game and how it could motivate her to exercise with her child and makes the game more fun.

> P01: [Make the game says:] "You're beating your mom today!" Or yeah, "You did this many steps, and you outdid your mom!"
>
> […]
>
> I: How would that affect you?
>
> P01: It'll make me want to go beat him. [laughter]

Similar to P01, P09 imagined how a competition between her and her child would be a fun way to assess their PA level, as it potentially spark a discussion related to health.

> P09: It's fun [to have a competition]. Fun competition to see how much you've done of the day. Like my daughter and I, and my daughter and I might say, "*Okay, which one of us were [higher]?*"

P16 also shared a similar positive perception about the "little competition" between parent-child. Later in the interview, P16 reconsider his use of the term "competition". This suggests that "little competition" may be different to other-referenced competition. Furthermore P16 saw the use of shared PA reflection experience as a source to spark health-centered discussions.

> I: Were you able to see how [child name] reacted when she saw the [SL] dashboard?
>
> P16: She was happy [to see her fuels go up in the SL dashboard]. It's like a little competition between us.
>
> I: Are you comfortable with a competition with your child?
>
> P16: It doesn't bother me, we just … we don't really have to make it like a competition, you know, she just be happy to get to the Open Gym, Saturday, to really see how much [spaceship] fuel she got. At the end of the, at end of the Open Gym, you know, we just could sit back and discuss how much fuel she earned.

P06 also described that parent-child competition could be used as an opportunity to bond with the child as well influencing the child to be more active by giving motivation and building confidence.

> P06: You know, you can make a competition if you want. Or if your child is, needs more exercise and the adult doesn't need as much, you're kind of helping using this as a way to kind of bond with your kid. And build some confidence and encourage her to get in shape.

P04, who initially said she was not motivated by the game, saw the value of parent-child competition to motivate her.

> P04: I wouldn't say it motivated me. Um, I think it's interesting. And I like to look at it. And I keep an eye on it. But it didn't inspire me to do more exercise.
>
> […] I think it might motivate me to have like a little competition with her [the child] of who was more physically active.

The common themes in parent-child competition are discussion, bonding, and motivation. This suggests that some parents saw competition within SL as a catalyst of interactions with their child, and they expect the interactions will strengthen their bond as a family. Furthermore, this motivates the parents to be engaged with the SL, as the interaction and the bonding experiences fulfill their need to feel related to their child.

Intrafamily Sibling Competition

A more challenging form of *intra-family competition* is *sibling competition* in which siblings could potentially compete. P6 was hesitant to have intrafamily competition to be present by default within SL game. He argued that every family is different and competition may lead to motivational or demotivational effect on different families.

> I: How do you think it would affect you, your child, and your family by having competition?
>
> P06: It depends if the family is very close. Sometimes having a competition can be fun. If the family is not really competitive type of family it can sometimes draw some people away. Because they'll feel like, "*Oh, well. I don't want to bother doing this, because I'm not gonna win anyway*."
>
> […] Like I said, there's people [who] gonna feel bad. Especially little kids. Six, seven years old, they may say, "*Well, I'm not doing well. I don't want to do it anymore*."

P06 provided a suggestion that families should be able to explore and decide whether to have or not to have intra-competition within the game. P06 specifically said "they need to search and find out", suggesting that attitude towards competition may evolve as the game progresses and the parents needs to continuously *manage competition* and monitor the siblings reaction.

> P06: So it depends on the dynamic of the family. So I think if they want to do it. Then they need to search and find out within themselves is this something that everybody's gonna be committed to.

P06 also suggested that family should be able to tailor the competition to fit the unique dynamics within every family. Finally, he concluded with his concerns towards competition.

> P06: I think within the game there should be a choice. You know, do you want to have a competition between members of your family. Or you guys want to monitor and just kind of keep track of each one. Without really put it in a competitive type of stance.
>
> […] Because you know, everything doesn't have to be a competition. I think that's some of the problems that we have now. You know, everything's a competition.

Family-to-family competition

When I asked about the potential of *family-to-family competition*, the participants were not against of the concept, but a few expressed a slight cautiousness. Interfamily competition by itself could be between *familiar families* or *new acquaintance families*. Two participants (P01 and P03) who are sisters imagined how the game would be more fun if their families can compete.

> P01: Well, we can compete against families. I think that would make it more fun. And then, I mean, as long I guess sign a paper saying, "*No hard feelings.*"
>
> P03: We can play with different families or a household like. My fence is my household. I have three girls and one boy. She has her son and her boyfriend and then three. We can be competing against each other and come over to the house on like a Saturday or take turns and see who won or something.
>
> P01: Or Sunday, when it comes to the last day.

The rich illustration from P03 suggests a strong desire for family-to-family competition because it was seen as a catalyst of positive family-to-family interaction, where virtual in-game interactions are expanded with real-world interactions. P01, however, was slightly more cautious by suggesting that families should have "no hard feelings". While P01 shared P06's aspiration to *manage competition*, P01 articulated that it should be done before the game begin. This is different to P06 who saw that *managing competition* a continuous process.

In the context of competing with *new acquaintance family*, P02 assured that she is comfortable of *family-to-family competition* with *new acquaintance family*. She further valued the friendly competition opportunity because the main game task is not to outdo others but to gain the health benefits, which is a mutually shared goal.

> P02: No, I'm fine. I'm comfortable with it [competing with new family]. I mean it will just make it sort of a friendly competition. You'll be wanting to try to go farther than the other family. Which to me is a good competition, because it's about exercise activities. So anything that gets you to exercise more is a good thing

Then P02 imagined how the *family-to-family competition* would be like with *new acquaintance family*. She suggested that the new acquaintance family should share the similar characteristics to her routine and her household. Then P02 described her family's characteristic and emphasized that she would not prefer to be paired with a family who is more active than her.

> P02: I mean it'd be nice if there was some type of, um, like maybe another family where it's, you know, in my case a single parent and two kids or somehow we would get paired kind of like family types. But no, I think it'd, I think it'd be fine. Because I mean, obviously if it's somebody who exercises all the time, we never do.

When I asked P02 to imagine if there is feature that would let her to communicate with the other family, she described that she would compare her PA with the other family and discusses ideas on how to be more active. P02 goes further suggesting that from this communications, the two new acquaintance family could meet and exercise together.

> P02: Um, then you could kind of ask them, "*Oh, I see you're doing a lot more. On this certain day you did a lot. What was your activity?*" Then you could, you know, maybe get some ideas or something you could do. To be more active by asking this other family, "*What did you do on day two when I see you were very active?*" It might give you some ideas that you didn't think of things you could do.
>
> Even you may end up maybe doing it together or something or "I'm going to the park with my child Sunday." "Oh, why don't I join you." Give them an opportunity.

Similar to P02, P09 also saw the value of *family-to-family competition* and saw the possibility that the kids might experience a negative feeling if they didn't win. Then P09 assured that there is nothing wrong about losing.

> P09: That [family-to-family competition] would be interesting. Yeah. Like "*My family has to win. Let's go. Have everybody involved*". Uh, I don't see the affect right now. I couldn't tell you. Probably for the kids who maybe feel like, "*Well, they did better than us*." You know, they may feel down, but there is nothing to feel down about it.

Then, P09 explained why not winning a family-to-family competition should not discourage game participation by citing the value of *task mastery* rather than outdoing the other. P09 also mentioning the "life lesson" of losing, suggesting that the game experience may have the benefit to *transfer family values*.

> P09: Because life is not always about winning. It's just doing your best. So if we do our best, and someone else happens to do even better than us, then the next time we'll do better than them. You know, you got to learn to lose too. You're not gonna win everything in life, so, yeah. That would not affect them.

However, P16 shows indifferent attitude towards competition. When asked whether having a leaderboard would affect his motivation, P16 responded:

> P16: Not necessarily. Because I don't have time to go to the gym where other family might have time to go to the gym to burn even more calories. It wouldn't bother me. Like I said, we wouldn't need a contest in my family. We just do we do it on a daily basis, you know. We just had a, to walking or whatever we may be doing.

Similar to parent-and-child competition, the parents saw family-to-family competition is seen as catalyst of social interactions with to attain a healthy life. The parents also suggest the need to feel related with the other family, further reinforcing the notion that competition is desirable if the actors feel related to one another.

Competition as a Catalyst of Social Interactions

It is clear that participants expressed some concerns about competition. However, the desire for competition seems to outweigh the concerns. While it could be said that our participants are the competitive type individuals, their stories suggest that competition is seen a catalyst of interactions. Competition is not just seen as an activity of outdoing others, but also a catalyst to enrich the exergame by sparking discussions and real-world social activities. By looking deeper into the responses, specific forms of competition seem to be desirable to motivate participant's engagement to the game.

Family-to-family competition, either with *familiar family* or *new acquaintance family*, seems to be desirable if the families felt they share similar character to one another. Within the Self-Determination Theory (SDT), this could be associated as the need to fulfill *relatedness*. Relatedness—the need to be connected with others—is one of the three psychological needs that elevates individual's self determination (others being competence and autonomy). In family-to-family competition, P01 and P03 (who are sisters) expressed an enthusiasm in competing with each other. This suggests that they perceive their relatedness would have a positive impact on the motivational and enjoyment when using SL. Later in the interview, I also asked P01 and P03 whether she would be interested to compete with families that they don't know, and they also expressed the willingness to play with new-acquintance family:

P01: We can pick a random family. We can pick someone we know.

P03: Yeah.

P01: A family we know. It doesn't matter. It can be just…

P03: We can even play with people like, you know how you can play, I think football.

However, it should be noted that they first raised the enthusiasm towards familiar family competition, and then approving new-acquintance competition. This suggests that their closeness or the similarity of character as a family takes more precedence when selecting which family to play with. This is similar to P02's request to be paired with s family with the characteristics that matches her family. In addition, early in this chapter P16's stated the feeling of being indifferent towards inter-family competition because he felt that other family does not share the unique characteristics of his family. This suggests that sharing a similar character has an influence whether two family can feel connected, and the families perceive this important to make them feel motivated to play the game and see competition as a catalyst of social interactions.

Parent-child competition seems to be desirable as it seen as a source to spark and enrich parent-child interactions in the form of health-centered discussions. Therefore, the parental desire of competition may be associated to the need of relatedness with the child, and this is supported by the interview data. This further underlines the connection between relatedness and competition, specifically how

comparisons competition in may cause parent-child interactions that fulfills relatedness.

These responses raised the need to see competition as a multifaceted catalyst of social interactions that is varied by the actors involved. The findings suggests the value of designing *friendly competition* that fulfills game player's need of *relatedness* by guiding the comparisons in a competiton towards health centered interactions. Fulfilling the need of relatedness potentially elevates game player's self-determination to be engaged to the exergame. Responses from the parents also suggest that the end result of competition is not about beating others. P06, P09, and P16 stated that the end result is to gain a competence in attaining a healthy life. This may conclude that a friendly competition is also driven by the need of competence.

> P09: Because life is not always about winning. It's just doing your best.
>
> P16: [parent-child competition is useful] just encourage one of us to do the best we can.
>
> P06: Maybe it shouldn't be a competition, but just kind of a friendly competition. No winners or losers. Everyone wins just because you're competing and you get something [the healthier life]. [It is] some type of incentive for just being in it [competition].

Study on SDT and PA found that relatedness and competence is dominantly related to task-mastery climate rather than others-referenced competition climate (i.e. to beat others) [6]. This may be an indication that a *friendly competition* is a task-mastery climate in which the comparison in competition is seen as a path toward social interactions.

4.3.4 Psychological Needs in Exergame

As evident in the previous section, the desire for *friendly competition* seems to resonate throughout the responses, as it is likely to fulfill the psychological need of *relatedness*. As the themes continue to materialize, the concepts emerge to fit the three psychological needs in SDT: relatedness, competence, and autonomy Figure **10**.

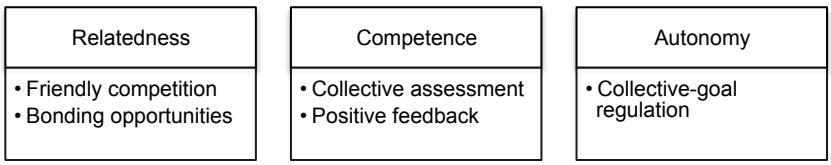

Figure 10 **The three psychological needs in family-based exergame**

The Need of Relatedness

A number of parents saw the game as a parent and child *bonding opportunity*, and thus fulfilling the need of *relatedness*, especially if they were not able to interact face-to-face regularly. In SDT, relatedness is the need to feel connected and be accepted by others.

> P06: Well, day to day, the kids are going to school. The adults are going to work. So there's really something that's not a lot of, *you know*, you're living your lives, so there's not a lot of, I mean for me personally speaking, there's not a lot of personal bonding all the time. So this is just another way to bond with your child.

In the quote above, P06 exemplifies the complex circumstances that families may face by describing how work, school, and routines constrained them to bond as a family. P06 further saw that the collaborative reflection interface in SL Dashboard as a supplement of the missing bonding opportunity.

One of our participant who is visually impaired, P08 had her daughter and her partner to read the numbers in the SL Dashboard. P08 described how the SL Dashboard helped her to know more about her and her child.

> P08: [The game is good] 'cause you know about your and your child's [physical activeness]. And you know both of them, not just one person. You know both of them, the kid and the parent.

Similarly, P09 described how SL could potentially give parents reason and motivation to be involved with their child.

> P09: [The game give a reason] for the parents, [so] they get involved with the kids, if they're not involved. And that [playing the game] would be a reason they get involved.

Accounts by P06, P08, and P09 indicated that SL is seen as a hub for family members to connect and to be involved. In effect, it fulfilled the need of relatedness, especially for families who unable to interact as frequent as they would like. In the previous section, *friendly competition* is also seen as a source to fulfill relatedness by sparking interactions. This suggests that fulfilling the need of relatedness by offering *friendly competition* and *bonding opportunities* leads participants to a stronger self-determination to engage with the game.

The Need of Competence

In the perspective of *competence*, the responses have suggested that *friendly competition* is potentially motivational if it is framed as an inspiration to the pursuit of *competence*. It could be accomplished by using *competence as benchmark*. For example, competition could be presented so that it inspires families to see the CDC's PA recommendation[3] is attainable. Therefore, individuals who completed the PA recommendation will not be seen as a winner that is exclusive to losers, but as an inspiration and a model to strive for.

Outside the context of competition, a number of parents described how they and their child collectively assess their competence to be more active. P01 described

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[3] CDC recommends 150 minutes of physical activity per week for adults

that having her and her child's PA level represented as a single entity in the SL Dashboard, she was compelled to bring her child to a *collective assessment* of PA.

> P01: After you see it [the game's dashboard] and you'd be like, "*Oh, it's not enough*." Or I'll be like, "*Oh, I did pretty good*." Or "*We both did good*."
>
> I'll be like, "Oh, look [child name], this is how much you did. And this is how much you did."

P09 also described her *collective assessment* experience, by detailing how she was in a *friendly competition* with her daughter. She described how she and her child assessed their numbers were going up and down, and valued the positive goal of the game. This implies that they were collectively assessing their PA in connection to their goal to have a healthy life.

> P09: Now we compete with each other to see which one of us has moved more. So some days she moves a lot more than me. And some days we move equal. Some days I move more. But all positive, no negative. It's the competition is not like in a negative way, you know.
>
> […] I think competitions good. With each other. As long as they're competing for positive things.

In contrast to P09, P08 described her experience of collective assessment as a cooperation that encourages parent and child to support each other by giving encouragement.

> P08: And that [observing the PA collectively] may encourage each other, we've got to do more or just like go keep doing the same thing we're doing right now. And [child name] actually did more this time.

These accounts suggest how the *collective assessment* feature in SL led some participants to continuously assess their competence collectively to meet the shared PA goal. By specifically bringing the collective aspect of it, the participants suggest that *collective assessment* provided a mechanics to fulfill the need competence and simultaneously satisfies the need of relatedness. Their appreciation provided support for the value of *collective assessment*.

Other parents saw the opportunity to use accomplishments in the game to motivate their child. P02 said that the SL Dashboard could be used to motivate the child, whereas P01 said that that the motivational elements should also be part of the game design elements.

> P02: I can also be able to show [child name], "*Look how much activity you did today.*"
>
> P01: Yeah, [give encouragement] like, "*Good job*."

P08 further identified the value of a motivational message to keep her child's self-esteem up. She also felt that at some extent the message is good for motivating the parents as well.

> P08: [It's good to have a motivational message within the game] to keep the child's self esteem up [by] saying they did a great job. […] It's good for the parents, but mainly for the child though.

P08 elaborated further how motivational message is good for children's self esteem, showing a determination to nurture as a mother.

> P08: Because they're [the children] the one that has to have the, you know, good self esteem. And that's good to tell them that anyway. That they did a great job. And they'll be more excited to begin [playing]. And you know, that way they can still continue to do, you know, what they like to do.

These accounts provided two ways in which positive motivational feedback could stem from the SL Dashboard: either initiated by the parents, or by the game. This highlighted how the parents perceive the importance of *positive feedback*, which in the SDT framework has been known to promote *competence*. Furthermore, the parents suggest the need for the SL game to confirm their child's need of competence, and at some extent: their own competence.

The mechanics to provide positive feedback is further materialized by P06 who proposed a detailed mechanism for an encouragement system so that the parents could send *positive feedback* when their child is meeting the PA target.

> P06: [In the game] you can let each person know. Like I said, everybody's busy. So you can text and say, "*Hey, so you reached your goal this week. Keep up the good work,"* or whatever. Or um if there was some type of, um, mechanism where if the person, if they've set a goal, the Fitbit can send a text message saying, "*Hey, you reached your goal this week. Congratulations. You are ahead of your mother*." Or "*You're ahead of your father in reaching your goals.*"

However, it is inevitable that some game player could fail meet the Spaceship Launch's goal. The value contextual information to overcome the feeling of "failing" was highlighted by P08. She suggested that there should be weather labels on each day on the SL Dashboard, in the form of sun or cloud pictogram. She suggests that it would help her to reason why she could not be active on certain day.

> P08: [The label is] for if you can't get out those days. The sun is for you're able to get out and do more physical activity. When the weather's snowy and rainy and stuff outside, you really can't do that much physical activity. Because of the weather.

She suggested other types of contextual information, such as when her child is sick or when the parent is tired. This suggests that when indivuals are assessing their competence it is also beneficial to acknowledge their limitations. Weather, illness, or stresses can restrict one's ability to be physically active, and having this barrier recognized within the SL Dashboard may be beneficial to overcome the negative feeling of not meeting the goal.

These findings suggests that the *collective assessment* in SL Dashboard led the parents to fulfill the need of competence by continuously assessing their performance collectively with their child. This further suggests that the *collective assessment* is also associated with the need of relatedness. The parents also talked about seeking cues of accomplishments to be used as a *positive feedback* to motivate their child, which suggest the parental desire to fulfill the child's need to feel competent.

The Need of Autonomy

The SL game is designed to encourage a team of parent and child to meet a *collective goal*. It appears that the motivational aspect of a collective goal is present in the collaborative game. P09 described how she motivated her child to meet the collective goal.

> P09: So I told her, I said, "*You went off to do a lot of activity and so we both can combine together and fill it up*." So it [the fuel goal] gives her a reason to look forward and gives me a reason to keep on moving.

Similar to P09, P03 also suggested the collective goal would motivate her child to be active and in effect would motivate the parent to be active as well.

> P03: But the kids gonna want it [to play the game] more. And the parents are gonna have to engage more into to, so it's not like the kids playing a game all day by itself or whatever.

The collaborative game also inspired P06 to co-regulate their PA goal. The desire for *collective-goal regulation* suggests the need of *autonomy* and *relatedness* at the same time. The need of *autonomy* is described as the need to in control over the outcome of individual's actions.

> P06: I think that's [exergame for a team of parent and child] an excellent idea. Um, because you can get, you can get your kids to, *you know*, you can set some goals together. And maybe even set the same goal.

P06's statement also highlighted the fact that every individuals are unique and a uniform goal might not match every individuals' capacity. When I asked P03 whether the game was too easy for her, P03 pointed out that indeed the game's challenge was too easy because of her active routine.

> P03: [Meeting the game's target] was easy […] because everything, *um*, things I have to do.

These statements may suggests the need to fulfill autonomy by providing the feature for families to customize the SL game's goal so that it fits to the uniqueness of every family.

4.4 Summary

Using SDT as a framework, I was able to see the relationship between the emerging themes from the field study and identified factors that encourage and inhibit individuals' engagement in the game. When I mapped these factors using the three psychological needs in SDT, it appeared that some these factors are related to one another.

Relatedness has been viewed as the most determining factor in PA self-determination [6]. Within SDT, relatedness is described as one's feeling to feel connected and be accepted by their social environment. The SL field study suggests that a *friendly competition* would promote *relatedness* because it is seen as a catalyst of interactions. Parent-child interactions are valued as a *bonding opportunity* to fulfill the

need to feel related. Participants also expressed a positive attitude towards *family-to-family competition* if the competition is paired between *familiar families* or *new acquaintance families* with a matching character (i.e. similar structure, routine, challenges). This suggests that for *friendly competition* to occur, the actors should feel connected to each other. In effect, friendly competition fulfills the need of relatedness and stimulates social interactions.

However, it is still unknown how sibling-competition would affect game engagement in *intrafamily-competition*, as one participant was concerned about the possibility of negative outcomes.

Competence, or one's perceived capability to complete a task, seems to be stimulated by *collective assessment* and reinforced by *positive feedback*. Collective assessment fulfilled the parental need of competence and relatedness as evident in how they continuously assessing their performance collectively with their child. The parents seeking cues of accomplishment to be used as a *positive feedback* to their child, suggesting the parental need to ensure their child to feel competent.

Autonomy, the need to feel a control over the outcome of one's action, may be to be related to *collective*-goal regulation. One participant described the need of this. Furthermore, the collective experience seems to suggest that the sense of *autonomy* could be supported by *relatedness*.

Accounts from the participants suggest that the SL Dashboard is increasing their PA intention. This intention is often translated into physical activity individually or with their kid. In some cases, the parents are generating PA ideas to get spaceship fuels.

In addition, having a PA monitoring and visualization unravel numerous *PA learning experience* among participants, such as remembering, understanding, and analyzing. Some participants was able to use SL Dashboard remember their physical activities. Other participants expressed an understanding about the difference between one type of activity to another by making comparisons. However, with the active scaffolding on concrete PA experience, the participants exhibited the analysis between PA and food choices. Furthermore, one participant expressed a conflict between her perception and the measured value of the caloric contents of a banana. This suggests that the learning experience may happen during game.

In conclusion, participants' accounts and suggestions provided insights how an exergame could be designed so that the engagement is more lasting. The findings support the value of satisfying the psychological need of *relatedness*, *competence*, and *autonomy* to promote participants' self-determination to engage with the exergame. Relatedness may be supported by designing *friendly competition* and providing *bonding opportunities*. Competence may be stimulated using *collective assessment* with *competence benchmark*, and reinforced by *positive feedback*. Autonomy may be fulfilled with the *collective-goal regulation*. The presence of learning experiences during the

gameplay suggests the opportunity to *actively scaffold* a better understanding of healthy lifestyle while the exergame's gamefulness and playfullness have not worn off.

Chapter 5
Discussion

Work in HCI has raised concerns over competition in exergame or gamified PA visualization [45,47,96]. This stems from the notion that the entertainment elements of exergames should never produce any negative nor unsustainable outcome [65]. However, results from the field study suggest that competition is not always undesirable. Stories from Spaceship Launch participants helped this work distinguishes the various kind of in-game competition, and which competitions are perceived to be useful to maintain PA and exergame's engagement.

Intrafamily competition is perceived to be positive when the comparisons are in the spirit of being connected as a family with a main purpose of being healthy. The participants saw that a competition will stimulate health centered discussions and physical activities in the real world. The endeavor to be healthy seems to outshine the negative feeling that may arise during comparisons. From the perspective of Self-Determination Theory (SDT), competition allows them to connect and fulfill their psychological needs of relatedness and competence at the same time. Indeed, a research in PA has shown that social systems that value efforts rather than comparisons are more likely to satisfy the three psychological needs in SDT: relatedness, competence, and autonomy [6].

While the parents show no aversion towards *parent-child competition*, one parent raised concerns about *siblings competition*. Furthermore, the same parent underlined that family dynamics may determine whether sibling-competition will lead to negative outcome or not. Family dynamics could be translated as the relationship between family members and how the family system governs these relationships. Family that values individual's victories (over the endeavor of a healthy life) will differentiate losers from winners. As a result, members of this family would feel less related to one another when playing such game, and individuals' self determination to play would be diminished. Therefore it is worth to pinpoint that the environment in which the competition took place may be equally important as the relationship of the actors.

Positive attitude is also shown towards *family-to-family competition*, under the condition if it is paired between *familiar families* such as an extended family. Interestingly, competition with *new-acquaintance family* is also desirable if the other family shared a similar structure, routine, and limitations. This suggests that fulfilling the need to feel related elevates the determination to be engaged with the game.

Taking a cue from the concerns over sibling-competition, family-to-family may also be impoverished if the environment put too much emphasis on egoistic victories. This suggests that competition in exergame is best facilitated by the presence of relatedness, competence, and task-mastery environment.

When it is present, relatedness is the social capital that contributes to the motivation to be engaged with a game. Competence is the in-game goal that the game player must accomplish. Task mastery is the environment that nurtures relatedness by valuing competence over outdoing others. The design implication is the game should recognize each family member's efforts as an endeavor to be healthy and inspire that healthy living is attainable. Competition is not always undesireable, but efforts should be made to frame competition as a *friendly competition* with a primary purpose of stimulating social interactions. Figure 11 visualizes this relationship into a conceptual model which I will discuss in the next section.

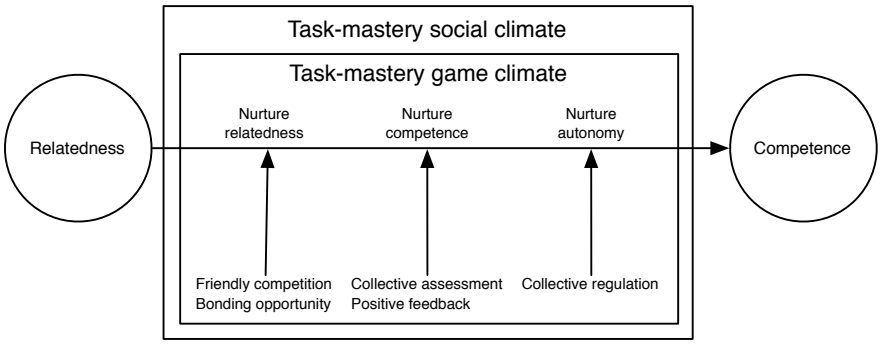

Figure 11 Task-mastery conceptual model for exergame in family setting

5.1 Task-mastery model for exergame

Using the SDT as the theoretical grounding [71] together with my study findings, I devised a task-mastery conceptual model for designing exergame for promoting physical activity in families (Figure 11). In this model, the game engagement is driven by two factors. First, the game's end goal of being competent in attaining an active life. Second, the satisfaction of relatedness between family members that that was present before and during the game play. This is supported by the work of Murcia that concluded physical activity enjoyment and motivation is fostered by the three psychological needs of competence and relatedness by the means of task-mastery climate [6]. Work by Peng et. al. on exergame also provided an evidence that game player's engagement is supported by fulfilling the need of competence and autonomy [62].

However, a task-mastery game relies to the larger system in which the game resides: the family system or the interfamily system. In Figure 11, this is depicted as "Task mastery social system". If the family put more emphasis on outdoing others rather than mastery, then the task-mastery game climate will be weakened by the larger system.

Data from the field study provided five design themes that may support relatedness, competence, and autonomy: friendly competition, bonding opportunities, collective assessment, positive feedback, and collective-goal regulation.

In a *friendly competition* comparisons between players should not be the end goal. Rather, comparison is just a path to spark discussions and activities that fulfills the need of relatedness. In-game comparison of one player to another should be designed to guide the player to discuss health implications and barriers. In effect, in-game competition would not just stop at comparisons, but should be extended to interactions that fulfill the need of relatedness and competence. In addition, it may be beneficial if the game provided activity ideas or encourage players to discuss activity ideas. This is important to guide the users towards social interactions promote relatedness. When a leaderboard design element is used, the leaderboard should convey the goal is to meet the recommended PA level, rather than ranking who exercised the most.

Bonding opportunity is related to *positive feedback*. Tracking devices allows continuous 24-hour a day recording of PA. Presenting family members' time series side by side allows family members to understand their family routine and in effect may afford family to form a stronger bond. Such system carries the characteristics of a socially translucenct system that supports visibility, awareness, and accountability. In a socially translucent system, individuals become aware of their actions because socially significant data is visible to everyone, and as a result they felt accountable for actions they choose [31]. In a family-based PA information system that is socially translucent, family members may become aware that the physical activities are visible to others, and they are held accountable for every activity and inactivity. For example a parent might become encouraged to exercise during lunch hour, so that their children can see that they are also physically active. Or a parent who mended household's work during the day may feel the effort is recognized because the children and the partner would be able to see his/her activeness. A teenager could use the activeness in the timeline as a proof to show that he/she was out with her friends for a walk. Family members could also use the PA timeline as the basis to send positive motivational feedback. For example, when the PA timeline is high, family members could send motivational praises. When the numbers are low, family could send a caring message and offer support.

This however raises a concern of privacy. Are teenagers comfortable to have the parent seeing their daily activity? Or whether the parent is comfortable to have the entire family member see their activity? While the purpose of this collective monitoring is healthy living, by observing a PA timeline one could infer what kind of activity it is. This defies the original purpose of a system that seeks the competence in being active and to feel connected, to a physical activity surveillance system. Would people remain to feel autonomous as an individual if the system becomes a surveillance system? One open question is how to balance the opportunity of collective PA monitoring (as a source of support and nurture) with the privacy concerns that may arise.

Collective assessment satisfies the need of relatedness when the parents are assessing their game's progress with their child to fulfill their need of competence.

This is related to *collective-goal regulation*, which is associated with the need of autonomy. In SDT, autonomy is defined as individuals' perception of full control over the outcome of one's action. For exergame designers, defining a goal is challenging because they are facing two opposing requirements: (1) fulfilling a standardized PA requirement from health agencies, and (2) acknowledging that every individuals' capability and circumstances are unique. Setting the goal too easy, the participant may not be motivated because of the lack of optimal challenge. Having it too difficult, some participants might become demotivated sooner and some may exercised too much. As I iterated before, exergames—as a serious game—are burdened with the responsibility not to cause any negative outcome [65].

Having an open level system that a user can freely choose may be useful to lessen the problem of standardized goal in exergames. It may also fulfills the need of autonomy, as one parent requested a feature to tailor the game to fit the uniqueness of every family. For example, an exergame could be designed to allow game players to set goal to be active 100 minutes a week (as opposed to CDC's 150 minutes a week recommendation). However, this also raises another question: should an exergame with a purpose of PA promotion allow individuals to set goals that does not meet health agency's recommendation? While an open level system (i.e. where people can actually choose the levels they like) may lead participants to choose levels that they can easily accomplish, it has the merit to provide a route and inspiration towards a healthier life and at the same time allows setbacks when individuals efforts are constrained by internal and external factors. Future work should devise an interaction model of goal-autonomy that directs game players toward the recommended health metrics.

This study provided support on designing task-mastery exergames that maintain engagement and enjoyment by fulfilling the need of relatedness, competence, and autonomy. This work also provided support for in-game competition under the condition that it is designed to spark discussions and interactions. However, it is still unknown how the social translucency that is inherent in this system could affect game player's sense of privacy. More work is also needed how game players could balance the PA goal while at the same time directing them towards a healthier life.

5.2 Harnessing the Learning Experiences

The results from the study suggest that several learning experiences emerged when our participants were using the SL Dashboard. Using Bloom's revised taxonomy of learning objectives, the emerging experiences could be categorized into remembering, understanding, and analyzing. One participant also reported a conflict between her old knowledge and the new knowledge. The presence of these learning experiences suggests the value of designing technology-based intervention with specific learning outcomes. In addition, the availability of concrete PA data is extremely useful to aid experiential learning.

The active scaffolding in Spaceship Launch could be expanded further by actively asking participants to compare one day to another with a purpose of identifying what kind of activity results in more moderate or vigorous PA. Recent smartphone applications are also capable to measure heart rate, which is an opportunity to scaffold the relationship between physical activity and heart rate. Indeed, novel health seniors open new opportunities for health-centered technology-based experiential learning system with a defined set of outcomes. However, future research should look into the challenge in scaffolding health-literacy using interactive computing.

In this study, one participant expressed a conflict between old knowledge and new knowledge. Indeed, learning is also an experience of unlearning previous knowledge. Kolb [40] highlighted that learners—in this context are the individuals who need the PA literacy—entered the learning experience with their own understanding of PA. Conflicts may emerge as the old knowledge crosses path with the new knowledge. In conventional learning, it is the role of an instructor to aid the learner to reconcile this conflict. On a computer aided learning, this raises a question how can computer understand this conflict and assist the learner to reconcile through health-centered dialectics?

One direction to pursue is guiding the learner by asking follow up questions during the reflective observation and abstract conceptualization phase. The purpose of these questions is to engage user in a dialectic assessment of their physical activity episodes. The questions could be framed as "*Is this accurate to your perception? Why do you think this is accurate/not accurate to your perception?*" or "*How do you think you could match your PA to burn specific food?*". However, since the variety of responses is likely to be high, interactive systems should be designed to ensure that the outcome aligns with the health recommendation. In addition, a conversational system may be needed to aid this dialogue. Therefore, more research is needed to see how interactive computing system could use the responses to these questions to scaffold a precise understanding of health.

Additional research direction is to go further into the experiential learning cycle by encouraging game's participants to experiment with physical activity intensity and duration. Having this feature will complete a single experiential learning cycle of concrete experience, reflective observation, abstract conceptualization, and active experimentation. Research in this direction could potentially uncover the full potential of experiential learning using PA tracking devices. It should be noted that this may present two opposing exergame's goal: (1) to maximize physical activity level to get most game points, and (2) to decrease and increase physical activity duration or intensity to see the its' relationship with health. It may be practical to have one exergame's goal, however it may be possible to intertwine the goals so that the two objectives could be accomplished in a single exergame.

Exploring abstract conceptualization and active experimentation within the framework of experiential learning is a promising research direction in the field of HCI, given the availability of health sensors that could provide a variety of data to reflect upon. The open research question is to understand how to design systems such that the health literacy could be precisely scaffolded.

5.3 Exergames for Lower-income Families

Childhood obesity has been linked to television viewing [21] and video games [93]. Study by Roberts [67] found that children of lower income families spent more time exposed to television and video games. The same study also found that parental education is a contributing factor in children's exposure to television and video games. Therefore, a well-designed exergame could potentially take back the sedentary television viewing time and supplant it with physical activities. Spaceship Launch provided a case study on deploying family-based exergame in a lower income community. However, more study is needed to uncover how exergames can engage children in longer period of time per day.

One potential direction to maintain engagement is by introducing social interactions within the game so that the children could play together remotely. This may be beneficial for families who perceive their neighborhood to be unsafe as it prevents them to play outdoor [51] which is suggested by the finding in the formative study. It should be noted that exergame play may be more active than passive television viewing and it may require more parental supervision. Our formative study found that some parents sometime handed over the role of babysitting to smartphones and tablets. Therefore, research is needed to understand how parents could let their kids engaged to a physically active game with a level of supervision similar to television watching or video game playing.

In terms of technical deployment, Spaceship Launch is hampered by a number of technical issues as presented in Table 8. The most prevalent problem is Fitbit wireless sync setup and followed with Fitbits being accidentally disconnected from the game. In addition, two families do not have computer at home, and as a result one of them decided to drop out from the study. While computer literacy may not be a problem that is limited to lower income families, the complexity level of Fitbit setup is the initial barrier to participation to exergame of this type. Therefore having a public display at community centers could provide some flexibility of access to families with limited computer literacy or families with no computer at home. This is supported by the finding by Roberts [67], where children of lower income families spent less time with computer compared to those of higher income families, which suggests lesser computer ownership among lower income families. Pew Research also reported that lower education attainment and lower income level is strongly correlated to internet usage [101]. Computer literacy concerns accounted 32% of the reason why adults chose not to use the internet, whereas the cost of internet and no-access accounted 19% and 7% respectively. This provided support for

multi modal deployment so that families could have the flexibility of access, either using personal devices or public access system deployed at community centers.

5.4 Summary

The social capital within families provided a starting point to deploy technology-based interventions for kids and parent. In the context of HCI, this work provided support to existing research [62,85] on the value of using Self-Determination Theory as a framework to design exergame with a more lasting enjoyment and motivation. This work also informs future work about the value of designing technology-based PA intervention as learning experiences with specific outcomes. However, HCI designer should also pay attention to the learning outcomes and how to ensure that they always aligned with the health recommendations. HCI designer should also look how privacy issues will not overshadow the social translucency aspect of social health monitoring. And last but not least, future work should explore how in-game goals in technology-based question can be designed to meet the health recommendation and at the same time acknowledging the uniqueness and challenges of every individuals.

Chapter 6
Conclusion and Future Work

The aim of this thesis work is to answer my overarching question of *how parent-child interaction on a technology-based intervention can facilitate health promotion and what are the design guidelines for creating such interventions*. I identified common concerns over the use technology for promoting physical activity, especially in family setting (Chapter 2). My work included a formative study (Chapter 3) to understand the requirements to design such technology in lower-income families, which is my target demography. To elicit the characteristics and requirements of such technology, my research partners and I conducted focus group interviews and surveys. Then I proceed with designing and evaluating Spaceship Launch, an exergame for parent and child (Chapter 4). I evaluated the system with using variety of methods (including interviews, participatory design workshop, surveys, log analysis) to understand participants' experiences and needs for an enjoyable and motivational exergames in family setting. Finally I synthesized the findings of my study to inform future work in HCI when designing exergame and gamified system for physical activity promotion (Chapter 5). In this section, I will briefly answer my research questions and reflect deeper to the emerging themes in my work.

6.1 Answering Research Questions

RQ 1: What are the design opportunities for family-based technology interventions that promote physical activity?

In my first preliminary study I found that the parents are supporting their child to be physically active by looking for exercise opportunities and providing support. However survey data suggests that the parents did not provide much modelling to the child by participating in PA together. Study found that exercising together is challenging for families. This suggests a design opportunity to design an exergame that uses parent and child physical activity as part of the game mechanics. These designs encourage families to look their physical activeness as a single unit even if they are physically active at different time or place.

In the preliminary study I also found the presence of PA perception gap, where the parents perceived they are more physically active than the actual objective measurement. A design opportunity for this problem is by providing computer aided health education by leveraging the concrete PA data from activity monitoring devices. The learning experience should lead the individuals to understand and analyze the relationship between PA intensity and duration with health.

RQ 2: How can a family-based exergame encourage parental intention to be physically active? How can parental modelling of physical activity be supported and impacted by this exergame?

Having access to PA data seems to elevate parental intention to be active. The participants reported their intention to be more active, and some participants translated this intention to actions. The pre and post survey on physical activity intention indicated that the majority of the participants expressed an increase of PA intention. This suggests that the combination of using Spaceship Launch and wearing the Fitbit had a some positive effect on physical activity. However, more work needs to be done to understand the extent and the sustainability of this effect.

In terms of PA modelling, only one participant reported to exercise with their child. However the desire for parent-child competition expressed by the majority of the participant suggests that PA modelling (i.e. parent exercising with the child) may happen if the game is designed as parent and child competition. The need of relatedness that resonates throughout the participants' stories suggests that the presence of child on exergame is a positive incentive to the engagement to exergame.

RQ 3: What types of interactions emerge in a collaborative exergame designed for intrafamily use?

The parents used Spaceship Launch to inspire their children to be active. Few parents used competition as an opportunity to connect and to motivate their child and themselves, even if the game is not designed for competition. Few others imagined that parent and child competition is also useful to spark health related discussion and real world activities. Some parents discussed their game results with a purpose to assess their physical activity level. One parent used PA visualization to give positive feedback for their child.

Accounts from the parents expressed their need to feel connected with their child by having conversation and activity related to health. They also expressed their desire to continually assess their competence to accomplish PA goal with their child. This supports for the value of fulfilling the need of relatedness to improve the enjoyment and motivation in family-centered exergame.

6.2 Limitations

This work is limited to see the impact of interaction of family-based exergame of two-member team. Therefore, this work has not studied a case study where the entire family members work together as a team. The active scaffolding of PA literacy through the use of trivia is also limited to the way the trivia is presented. Currently the trivia is presented the first time of the day a user accessed the Spaceship Launch dashboard. If the user accessed the Dashboard in the morning and he/she was not active yet, the user may not receive any trivia or the trivia is based on a very light activity.

6.3 Future Work

This study provided support on the use of Self-Determination Theory to design exergames. Further research needs to be done to identify game design elements that fulfill the need of relatedness, competence, and autonomy. Moreover,

the emerging learning experiences suggests the need to put more emphasis on delivering health-based learning as an alternative to address the concern of novelty effect. The value of relatedness is a resonant theme among the participants, which advances the HCI research frontier on the use of technology-based PA interventions in family setting. Future work should conduct larger, longitudinal studies to further validate these findings and examine how such game are used over an extended period of time.

6.4 Concluding Remark

The affinity between family members is a valuable social capital for technology-based intervention in family setting as it fulfills the need of relatedness. In the scope of HCI and game design, this work highlights the need to foster this social capital by valuing the need of being healthy over the competition of outdoing others. This work distinguishes several kinds of competition, and competition that is designed as a path towards health-centered conversations and activities may satisfy family's need of relatedness.

Appendix A
Research Instruments for the Initial Formative Study

A.1 Interview Guide

Opener (3 minutes)

Please state your name, how long you have been attending SOG, and the ages of the children you bring to SOG.

Social Support for Physical Activity (15 minutes)

There are many ways that people can provide support to one another. I'd like to talk about support (or lack thereof) amongst SOG families.

1. Do any of the **parents at SOG** currently support you in providing physical activity opportunities for your child? Examples of support are:
    - **assisting you in** providing physical activity opportunities (e.g., helping you plan activities, take your child to activities, etc.)?
    - **going with you to** provide physical activity opportunities for your child (e.g., going to the park, on a walk, etc.)?
    - **discussing your challenges or successes** around providing physical activity opportunities for your child?
    - sharing information about **physical activity opportunities** with you (*e.g.*, telling you about programs such as SOG, sports camps)?
    - *Probe: barriers to support/why support not provided?*
    - *Probe: how could support be improved?*
    - *Probe: tell me about the people you interact with the most (characteristics, e.g., similar aged kids, etc.)?*
    - *Probe: strengths/weaknesses of SOG support?*
    - *Probe: do you need/want support from SOG families?*
2. Are you aware of the SOG Facebook page?
    - How useful is the SOG Facebook page?
    - Is there anything that keeps you from looking at it?

- Is there anything that drives you to look at it?
- Have you ever considered posting something to the page (why/why not)?

Emotion & Physical Activity (10 minutes)

For 60 seconds, think about how you feel when you are physically active with your child. On your worksheet, write words that describe how you feel while you are doing that activity. The words can be positive, negative or somewhere in between.

If you can't recall a time when you have been physically active with your child, imagine how you think you would feel.

[after 60s] Circle the 1-2 words that best describe your experiences.

3. Tell me about the words you wrote down.
    - *Probe: Which words best describe your experiences?*
    - *Probe: can you discuss some times when you have felt that way? What was happening?*
    - *Probe: Can you tell me what leads you to feel that way?*
    - *If negative words, probe: anything that could be done to make the experience positive?*

Technology (7 minutes)

For 60 seconds, think about a recent time when you used a piece of technology with your child. Examples are computers, cell phones, tablets/iPads, electronic toys (like LeapFrog) and video game consoles. On your worksheet, write words that describe that experience.

[after 60s] Circle the 1-2 words that best describe your experiences.

4. Tell me about the words you wrote down and the technology you were using.
    - *If no words: Frustrating? Fun? Difficult? Tiring? Hilarious? Necessary?*
    - *Probe: Which words best describe your experience?*
    - *Probe: Can you tell me what leads you to feel that way?*
    - *Probe: technology being used by both or one watching while other operates?*
    - *Probe: is what you've described what normally happens when you use technology with your child?*
    - *If negative words, probe: anything that could be done to make the experience positive?*

Closing (3-5 minutes)

5. Is there anything else you wanted to say but did not have a chance to?

Thank you for your time and your valuable input. We hope that you will also participate in a **follow-up telephone survey** to help us learn a little bit more about your thoughts on physical activity, SOG, your neighborhood, and how you use technology. If you haven't signed up already, you can see one of us to do so. **In the Fall**, we will be conducting follow-up sessions where we will work together to brainstorm ways for technology can encourage physical activity. If you would be interested in participating or learning more about those sessions, please let one of us know.

A.2 Telephone Survey

Note to Survey Administrator: Text in brackets ([]) should not be read, these are instructions to you. Also, do not read section headers.

Introduction

Hello, my name is <name> and I am a student researcher at Northeastern University. May I please speak with <participant name>?I am calling to conduct the phone interview that you signed up for. Is now still a convenient time for you? I am going to ask you a series of questions. If my questions are unclear at any time, please let me know. This survey should take about 30 minutes to complete. I want to remind you that everything you say today is confidential. We ask that you please answer the questions honestly and truthfully.

Do you have any questions before we begin?

Section A: Cell Phone Use

I am going to ask you about how you use different features on your cell phone. The options will be: never, few times/year, few times/month, few times/week, one or more times/day, and I don't know.

How often do you… [check one answer for each question]

	Never	Few times per year	Few times per month	Few times per week	One or more times per day	I don't know
…send/receive text messages/ pictures on your cell phone?						
…browse the Internet on your cell phone (check one answer)?						
…play games on your cellphone?						
…use health-related applications on your cellphone?						

Section B: Saturday Open Gym

Now, I'd like to ask you what you think about Saturday Open Gym.

1. Is there anything that you like about Saturday Open Gym?
2. Is there anything that you would like to change about Saturday Open Gym?
3. Does SOG encourage you and your kids to be physically active during the week?

   ☐ Yes, every week ☐ Yes, some weeks ☐ No, never

   Why/why not?

4. Do you participate in physical activity at SOG? (Choose one answer)

   ☐ Yes, every week ☐ Yes, some weeks ☐ No, never

   Why/why not?
5. In the past month, how often have you communicated with SOG parents (in person, on the phone, or on the computer)? (Check one answer)

   ☐ Every week ☐ Some weeks ☐ Never
6. Please choose the statement that best describes you. (Circle one letter)
   a. I would like to have more interaction with SOG parents than I currently do.
   b. I am satisfied with the amount of interaction that I have with SOG parents.
   c. I would like to have less interaction with SOG parents than I currently do.

Section C: Parent Physical Activity

Physical Activity is any activity that increases your heart rate and makes you get out of breath some of the time. Some examples of physical activity are running, brisk walking, rollerblading, biking, dancing, and swimming.

I'd like to now ask you a few questions about your physical activity.

1. Are you currently thinking about **increasing** the amount of physical activity that you get?

   ☐ Yes ☐ No **[if No, skip to Question 3 below]**

   2. [If **Yes** for question 1] When do you plan to begin to get more physical activity?
      - ❑ Within the next month
      - ❑ Within the next 6 months
      - ❑ Yet to decide
   3. [If **Yes** for question 1] How **confident** are you that you will stick to this plan?
      - ❑ Very confident
      - ❑ Somewhat confident
      - ❑ Mildly confident
      - ❑ Not at all confident

4. Have you ever made a deliberate effort to increase the amount of physical activity that you get?

   ☐ Yes ☐ No **[if No, skip to Question 5 & 6 below]**

   5. [If **Yes** to question 4] Are you still getting more physical activity?

      ☐ Yes ☐ No

   6. [If **Yes** to question 4] – Have you been able to maintain this increased level of physical activity for 6 months or more?

      ☐ Yes ☐ No

7. Over the last 7 days, on how many days were you physically active for a total of at least 60 minutes per day? [circle one choice]

   0 1 2 3 4 5 6 7

8. Over a typical or usual week, on how many days are you physically active for a total of at least 60 minutes per day? [circle one choice]

   0 1 2 3 4 5 6 7

Section D: Physical Activity Enjoyment Scale

Can you remember any recent times when you have been physically active? (Circle One)

- ❑ Yes
- ❑ No

If No: Please imagine how you think you would feel while doing physical activity.

If Yes: Recall the last few times that you have been physically active.

Next, I would like you to rate how you feel about physical activity *at the moment you are doing it*. I will go through a list of feelings that relate to physical activity. On a scale of 1 to 7, for each feeling, please tell me the number that best describes you.

Overall, when you are doing physical activity, how do you feel about it on a scale of 1 to 7, where 1 is [feeling] and 7 is [feeling]?

	1	2	3	4	5	6	7	
I enjoy it								I hate it
I am very absorbed in this activity								I am not at all absorbed in this activity
I feel good physically while doing it								I feel bad physically while doing it
I find it energizing while doing it								I find it tiring while doing it
I am very frustrated by it								I am not at all frustrated by it
It gives me a strong sense of accomplishment								It does not give me any sense of accomplishment at all
It's very refreshing								It's not at all refreshing
I felt as though I would rather be doing something else								I felt as though there was nothing else I would rather be doing

Section E: Social Cohesion

How much do you agree or disagree with the following statements about adults attending

Saturday Open Gym? Please circle one number for each statement.

Do you…[read number labels, not numbers]

	1 strongly disagree	2 somewhat disagree	3 neither disagree	4 somewhat agree	5 strongly agree
1. People at Saturday Open Gym are willing to help	1	2	3	4	5
2. The adults who attend Saturday Open Gym are a close-knit group.	1	2	3	4	5
3. People at Saturday Open Gym can be trusted.	1	2	3	4	5
4. People at Saturday Open Gym generally don't get along with each other.	1	2	3	4	5
5. People at Saturday Open Gym do not share the same values.	1	2	3	4	5

Section F: Child Background Information

Many of the questions that I will ask in the rest of this survey are about your child's activities. When I say "your child", please think about the child who attends Saturday Open Gym (SOG).

Equipment Checklist

Next, I am going to ask you about the electronic equipment that your child has access to in your home. By "have access to" I mean devices that your child is allowed to use.

How many ___ does your child have access to in your home?	**Total Number**
1. TVs	
2. Computer with Internet access or without Internet Access	
3. Tablet computer (e.g., iPad)	
4. Video game player that hooks up to a TV (e.g., PlayStation, Xbox) List:	
5. Hand-held video games (e.g., Sony PSP, Nintendo DS)	
6. Cell phone	

Section G: Child Physical Activity

Please add up the time your child spends in physical activity each day (do not include school physical education or gym class). As I ask the next two questions, please provide the answer that best applies to your child.

1. For the past seven days, how many days was your child physically active for a total of at least 60 minutes per day?

   0 1 2 3 4 5 6 7

2. Over a typical or usual week on how many days is your child physically active for a total of at least 60 minutes per day?
   0 1 2 3 4 5 6 7

Section H: Social Support

I have a few more questions to ask that will help us learn a little about your child's physical activity.

In general, how often do you or your partner…

	Never				Daily
…tell your child that being physically active is good for his/her health?	1	2	3	4	5
…watch your child participate in physical activity or sport?	1	2	3	4	5
…encourage your child to do physical activities or play sports?	1	2	3	4	5
…do a physical activity or play sports with your child?	1	2	3	4	5
…provide transport so your child could go to a place where he/she can do physical activities or play sports?	1	2	3	4	5

Section I: Neighborhood Factors

Neighborhood Barriers

Next, I would like to ask you some questions about your neighborhood. We are interested in how neighborhood surroundings affect kids' ability to be physically active. Please say the answer that best applies to your child. Do you agree or disagree with the following statements:

It is difficult for my child to be active in the local park, street, alleyway, cul de sacs near our home because... [circle one]

Do you… [read number labels, not numbers]

	1 strongly disagree	2 somewhat disagree	3 somewhat agree	4 strongly agree
There is not enough space to be active in	1	2	3	4
There is no choice of activities	1	2	3	4
There is no equipment	1	2	3	4
There is no adult supervision	1	2	3	4
There are no other children there	1	2	3	4
It is not safe because of crime (strangers, gangs, drugs)	1	2	3	4
My child gets bullied, teased, harassed	1	2	3	4
It is not safe because it is close to a road	1	2	3	4
There are too many people there	1	2	3	4
It does not have good lighting	1	2	3	4
It is difficult to get to	1	2	3	4

Neighborhood Surroundings

We have a few more questions about your neighborhood. Please provide the answer that best applies to the neighborhood where you and your child live.

Do you… [read number labels, not numbers]

	1 strongly disagree	2 somewhat disagree	3 somewhat agree	4 strongly agree
There are sidewalks on most of the streets	1	2	3	4
There are many interesting things for my	1	2	3	4
There are many beautiful natural things for my child to look at in my neighborhood (e.g., gardens, views).	1	2	3	4
There are many buildings/homes in my neighborhood that are nice to look at for my child.	1	2	3	4

Conclusion

1. This is the end of the survey. Thank you very much for participating!
2. You will receive the $10 Target gift card for your participation. I would you like to mail the gift card to you,
    i. Can you please verify the spelling of your first and last name?
    ii. What mailing address should we use?
3. Do you have any questions?
4. Thank you very much for your time. Have a great [day/evening].

Appendix B
Research Instruments for Spaceship Launch Evaluation

B.1 Interview Guide

Thank you for participating in this interview. It should last in 45 minutes. We will discuss your experience with the game and ask you feedback on how the game could be improved to encourage family fitness.

Remember there are no right and wrong answer. Please answer the question as honestly as possible.

I would like to audio record this interview. This helps us to thoroughly study what our participants have said. Is it OK if I audio record this interview?

Great, I'm starting the recording now.

Experience with the virtual rewards.	*Every weekend you could launch your spaceships to planets in outer space. The number of planet you can travel depends on how active you and your child are.* 1. Tell me how was the experience when you have enough fuel to launch your spaceship. 2. Does it affect or doesn't affect your activity in the following week (did you feel encouraged, discouraged, or not affected)? • If yes: would you tell me examples how if affected you? • If no: why do you think the game did not affect you?
Physical activity awareness and intention	*You have been using the Spaceship Launch game for ___ weeks. Using the dashboard you were able to see how much time you spend on physical activity.* 3. Did the experience with the dashboard is affect or not affect the way you see your physical activity level? Tell me more. 4. Was the visualization close or not close to they way you see your activity level? 5. Did it motivate or didn't motivate your intention to be physically active? Would you tell me examples how it

	motivated/demotivated you?
Effect of contextual physical activity reflection	*Every evening the dashboard present you with a question about your physical activity level and foods. The question asks you to reflect back to you physical activity episode of that day.* How did the reflection of your past PA affect or not affect your perception of physical activity? Did you feel that you need to exercise less, more, or no change? Could you elaborate on that? *The trivia question also asked you to associate your physical activity with calories in food.* 1. Did this change or not change your perception of the nutritional values of food? 2. How did this change affect your intention to be physically active? 3. How did this change affect the way you select food for you and your family? Could you give me examples?
Parental modelling	*Using the dashboard you were able to see how much time you and your child spent in physical activity.* Tell me how the experience was? Did you check the dashboard with your child? ❑ If yes, did he/she react to it at all? o If yes, how was his/her reaction when he/she saw his/her and your physical activity levels? o If no, why do you think he/she did not react to the dashboard. ❑ If not, could you tell me why you did not to check your PA with your child? How was your experience of seeing your activity level with your child? Does monitoring you and your child's physical activity simultaneously impact your intention to be physically active. Could you elaborate on that.
Open Ended	Anything else you would like to say about your experience using Spaceship Launch?

B.2 Survey

Thank you for taking the time to take part in the study. This survey will take about 20 minutes to complete. There are 5 sections in 7 pages. Your response will be kept confidential. Instructions have been provided for each section. If you have any questions please ask our research staff for assistance

A. Physical Activity

Please complete the following sentence: "I intend to engage in physical activity during the next 4 weeks the following number of times:" (circle one answer below)

0	1-4	5-8	9-12	13-16	17-20	21+

Over the past 7 days, how would you rate your physical activity level? Circle one answer:

Low	Rather low	Intermediate	Rather high	High

How do you assess the *intensity* of your physical activity? *Intensity* refers to amount of effort required to do an activity.

Over the past 7 days, how many days were you physically active for a total of at least 30 minutes? Circle one choice:

1 2 3 4 5 6 7

Regular exercise is any *planned* physical activity (e.g., brisk walking, aerobics, jogging, bicycling, swimming, rowing, etc.) performed to increase physical fitness. Every week, adults need at least :

- 2 hours and 30 minutes (150 minutes) of moderate-intensity activity (e.g., brisk walking), OR
- 1 hour and 15 minutes (75 minutes) of vigorous-intensity activity (e.g., jogging or running)

Exercise does not have to be painful to be effective but should be done at a level that increases your breathing rate and causes you to break a sweat.

Do you exercise regularly according to that definition? (Mark one answer)

_____ Yes, I have been for MORE than 6 months.

_____ Yes, I have been for LESS than 6 months.

_____ No, but I intend to in the next 30 days.

_____ No, but I intend to in the next 6 months.

_____ No, and I do NOT intend to in the next 6 months.

B. Food Healthfulness

Estimate how many calories do you think are in each of these foods:

Food	Calorie
Salad (3/4 cup with 1 tbsp ranch dressing) lettuce, carrot and tomato	
Apple (2 apples)	
Black beans (1 cup canned)	
Banana (2 small bananas)	
Low fat yogurt (1 cup vanilla low fat)	
Fish (8.5 ounces; grilled ocean cod)	
Orange juice (Tropicana: 16 ounces)	
Baked chicken (7 ounce baked breast)	
Hershey's Kisses (4 pieces)	
Ice cream (2/3 a cup of Breyer's vanilla)	
Soda (16 ounce cola; non diet)	
French fries (McDonald's small fry)	
Cookies (4 Oreo cookies)	
Potato chips (1/2 ounce of Ruffles barbecue)	
Pizza (Pizza hut: 1 slice medium cheese pizza with thin crust)	
Hamburger (McDonald regular hamburger)	

How healthy are these foods? Circle your answer:

1 means very unhealthy and 8 means very healthy

	Very unhealthy							Very healthy
Salad (3/4 cup with 1 tbsp ranch dressing) lettuce, carrot and tomato	1	2	3	4	5	6	7	8
Apple (2 apples)	1	2	3	4	5	6	7	8
Black beans (1 cup canned)	1	2	3	4	5	6	7	8
Banana (2 small bananas)	1	2	3	4	5	6	7	8
Low fat yogurt (1 cup vanilla low fat)	1	2	3	4	5	6	7	8
Fish (8.5 ounces; grilled ocean cod)	1	2	3	4	5	6	7	8
Orange juice (Tropicana: 16 ounces)	1	2	3	4	5	6	7	8
Baked chicken (7 ounce baked breast)	1	2	3	4	5	6	7	8
Hershey's Kisses (4 pieces)	1	2	3	4	5	6	7	8
Ice cream (2/3 a cup of Breyer's vanilla)	1	2	3	4	5	6	7	8
Soda (16 ounce cola; non diet)	1	2	3	4	5	6	7	8
French fries (McDonald's small fry)	1	2	3	4	5	6	7	8
Cookies (4 oreo cookies)	1	2	3	4	5	6	7	8
Potato chips (1/2 ounce of Ruffles barbecue)	1	2	3	4	5	6	7	8
Pizza (Pizza hut: 1 slice medium cheese pizza with thin crust)	1	2	3	4	5	6	7	8
Hamburger (McDonald regular hamburger)	1	2	3	4	5	6	7	8

C. Parent and Child Physical Activity

During a typical week, how often do you: (circle one answer for each statement below)

1. Encourage your child to do physical activities or play sport?

None	Once	Sometimes	Almost every day	Everyday

2. Participate in physical activities or play sports with your child?

None	Once	Sometimes	Almost every day	Everyday

3. Accompany your child to go to places where he or she can do physical activities or play sports?

None	Once	Sometimes	Almost every day	Everyday

4. Watch your child participate in physical activities or sports?

None	Once	Sometimes	Almost every day	Everyday

5. Tell your child that physical activity is good for his or her health?

None	Once	Sometimes	Almost every day	Everyday

6. Talk to your child about physical activity or sports that you have done?

None	Once	Sometimes	Almost every day	Everyday

D. Families in Saturday Open Gym

How much do you agree or disagree with the following statements about families attending Saturday Open Gym? *Circle one answer for each statement.*

1. Families at Saturday Open Gym are willing to help each other.

Strongly disagree	Somewhat disagree	Neither disagree nor agree	Somewhat agree	Strongly agree

2. Families who attend Saturday Open Gym are a close-knit group.

Strongly disagree	Somewhat disagree	Neither disagree nor agree	Somewhat agree	Strongly agree

3. Families at Saturday Open Gym can be trusted.

Strongly disagree	Somewhat disagree	Neither disagree nor agree	Somewhat agree	Strongly agree

4. Families at Saturday Open Gym generally don't get along with each other.

Strongly disagree	Somewhat disagree	Neither disagree nor agree	Somewhat agree	Strongly agree

5. Families at Saturday Open Gym do not share the same values.

Strongly disagree	Somewhat disagree	Neither disagree nor agree	Somewhat agree	Strongly agree

E. Demographic Information

Which neighborhood of Boston do you live in? (Check one answer.)

___ Allston/Brighton
___ Beacon Hill/Back Bay/West End
___ Charlestown
___ Chinatown
___ Dorchester
___ East Boston
___ Fenway/Kenmore
___ Hyde Park
___ Jamaica Plain
___ Mattapan
___ Mission Hill
___ North End
___ Roslindale
___ Roxbury
___ South Boston
___ South End
___ West Roxbury
___ I am not sure
___ I do not live in the city of Boston
___ Other: _____

Do you consider yourself Hispanic or Latino? (Please check one answer.)

___ Yes

___ No

What is your race? (You can check more than one answer):

___ White
___ Black or African American
___ Asian
___ Native Hawaiian or Pacific Islander
___ American Indian or Alaskan Native
___ Other

How many children (under 18 years old) live in your household? ____ children

What is your marital status? Please check one:

___ Married

___ Single and never married

___ Widowed/divorced/separated

___ Living with partner

What is the education level you have completed? Please check one:

___ Less than 7th grade

___ Junior high/middle school

___ Some high school

___ Completed high school

___ Some college or vocational training

___ Completed college or university

___ Completed graduate degree

What is your approximate annual household income? Please check one:

___ < $10,000

___ $10,000 - $19,999

___ $20,000 - $29,999

___ $30,000 - $39,999

___ $40,000 - $49,999

___ $50,000 - $59,999

___ $60,000 - $69,999

___ $70,000 - $79,999

___ $80,000 - $89,000

___ $90,000 - $100,000

___ > $100,000

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